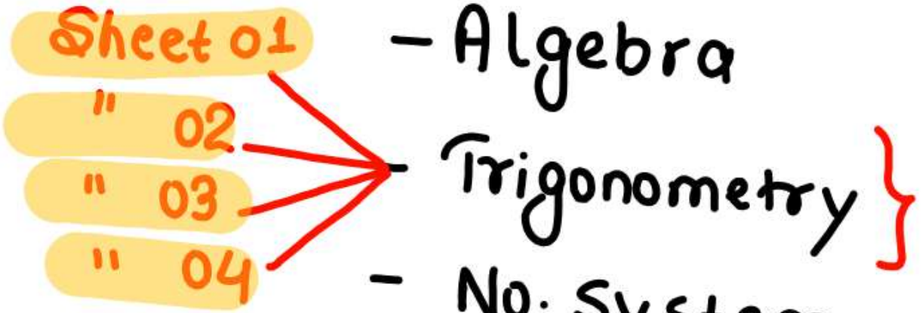
TRIGONOMETRY

त्रिकोणमिति

SHEET-01

CLASS NOTES

Advance Maths



- Algebra
- Trigonometry
- No. System
- Geo
- Mens

	Pre	Mains
Trigonometry	1-2	2-3

QUESTIONS BASED ON BASIC TRIGONOMETRIC RATIOS

T-Ratio

→ $\sin\theta$

→ $\cos\theta$

→ $\tan\theta$

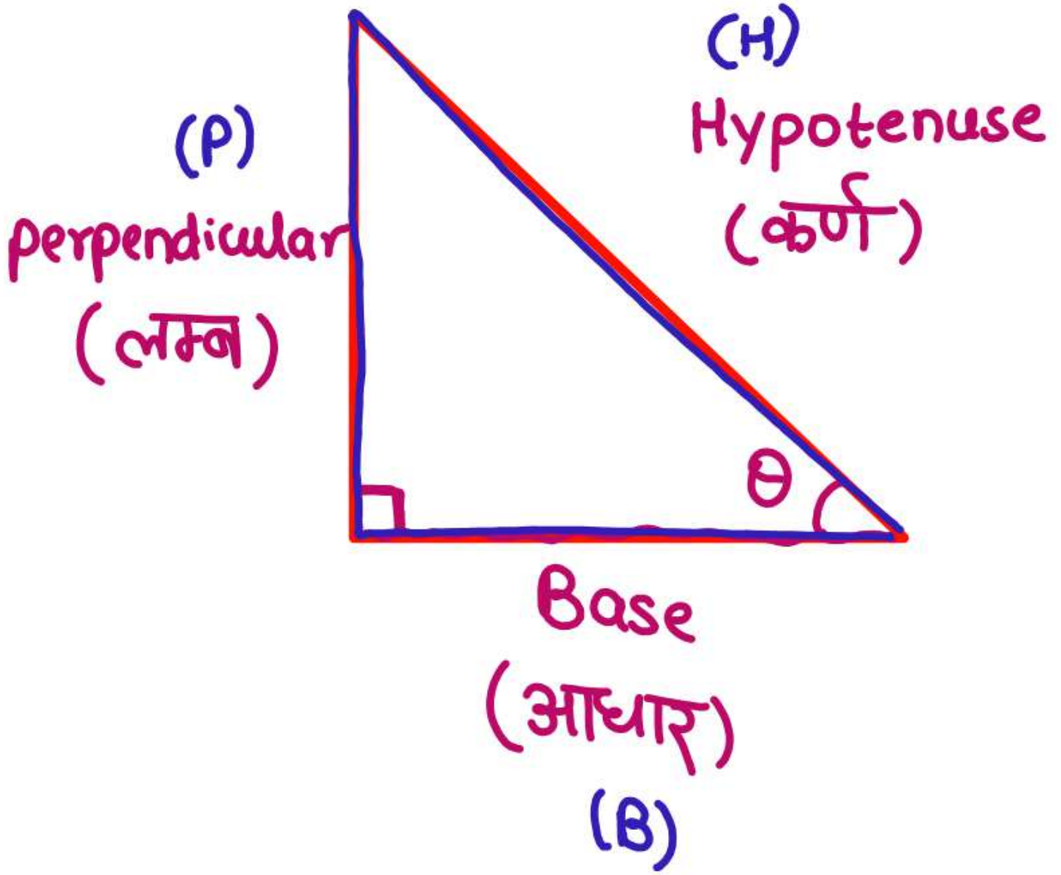
→ $\operatorname{cosec}\theta$

→ $\sec\theta$

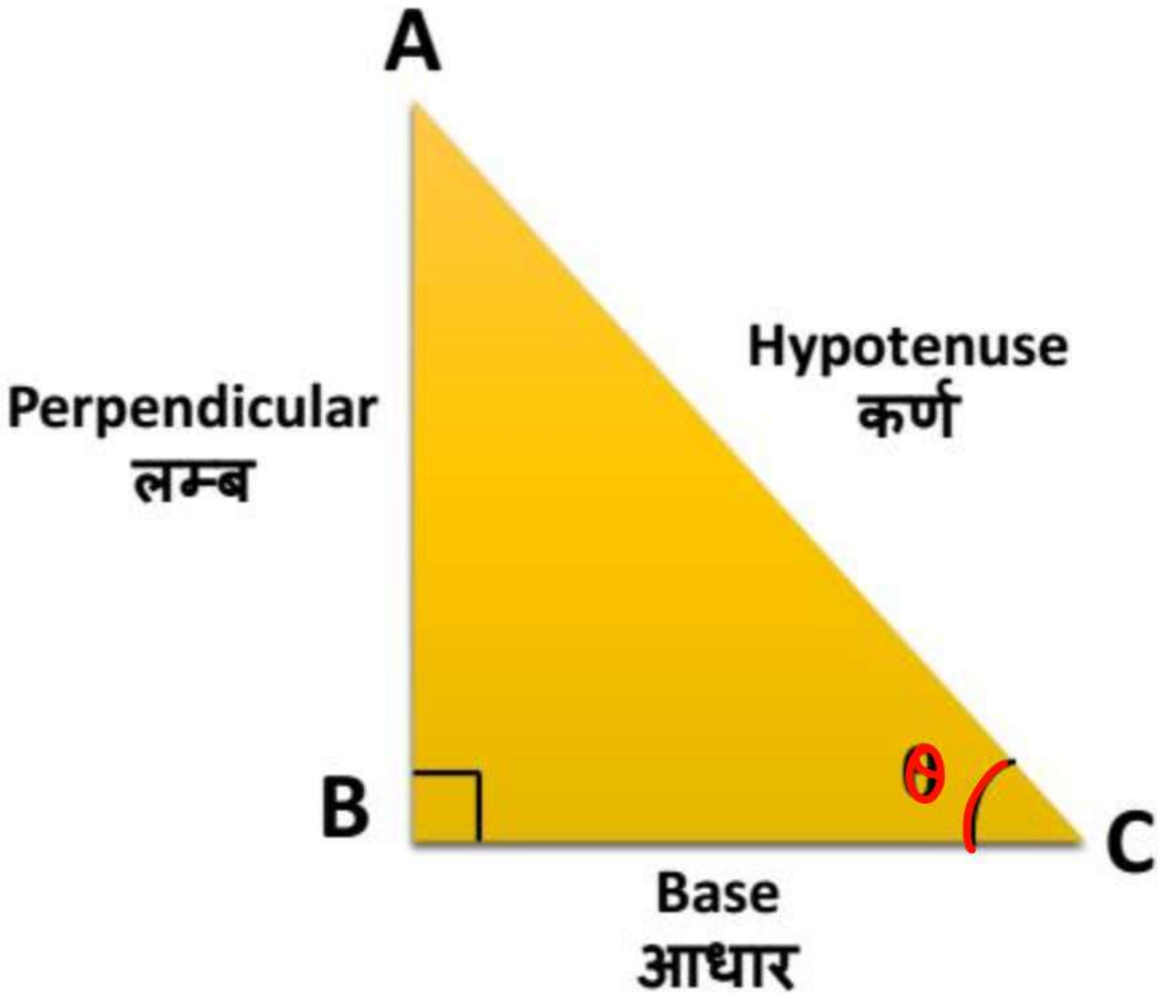
→ $\cot\theta$

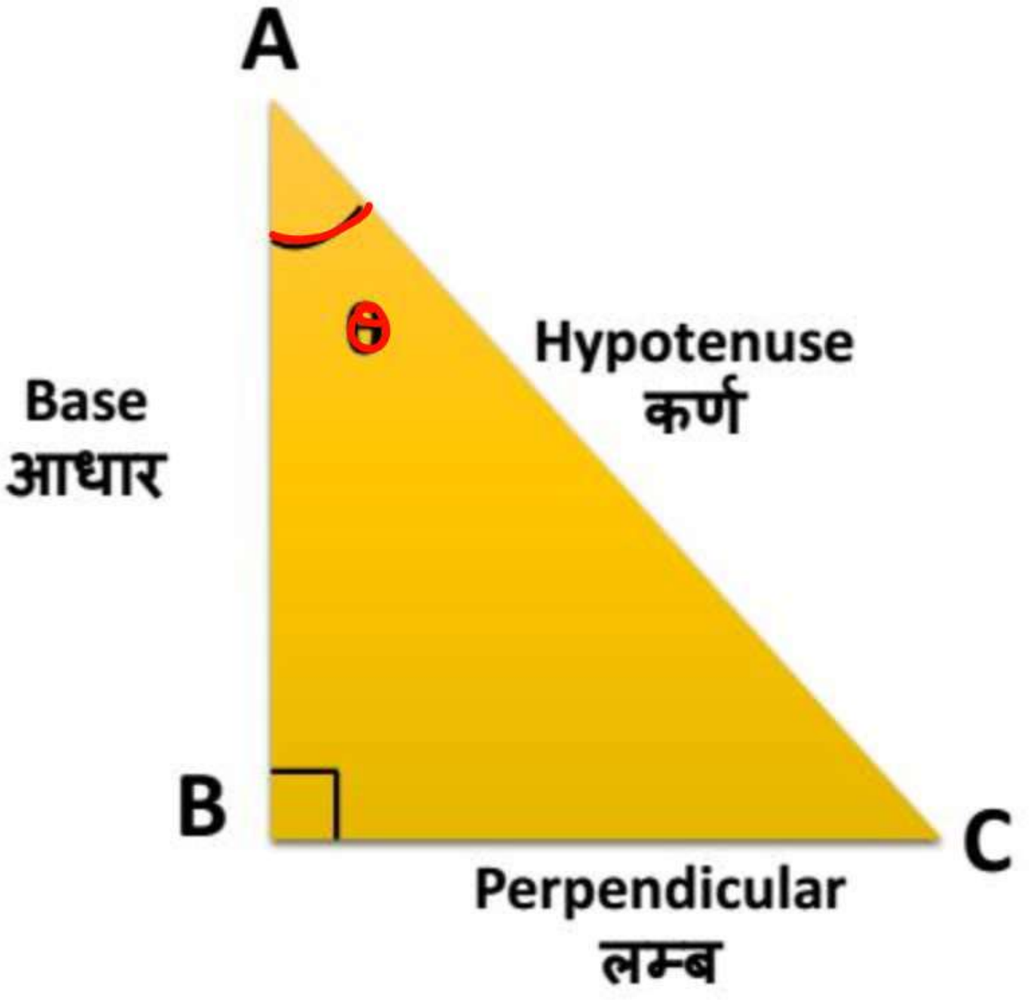
$\sin\theta = \frac{P}{H}$ $\cos\theta = \frac{B}{H}$ $\tan\theta = \frac{P}{B}$

$\text{cosec}\theta = \frac{H}{P}$ $\sec\theta = \frac{H}{B}$ $\cot\theta = \frac{B}{P}$



CONCEPT OF BASE AND PERPENDICULAR





HOW MANY TRIGONOMETRIC RATIOS ARE THERE?

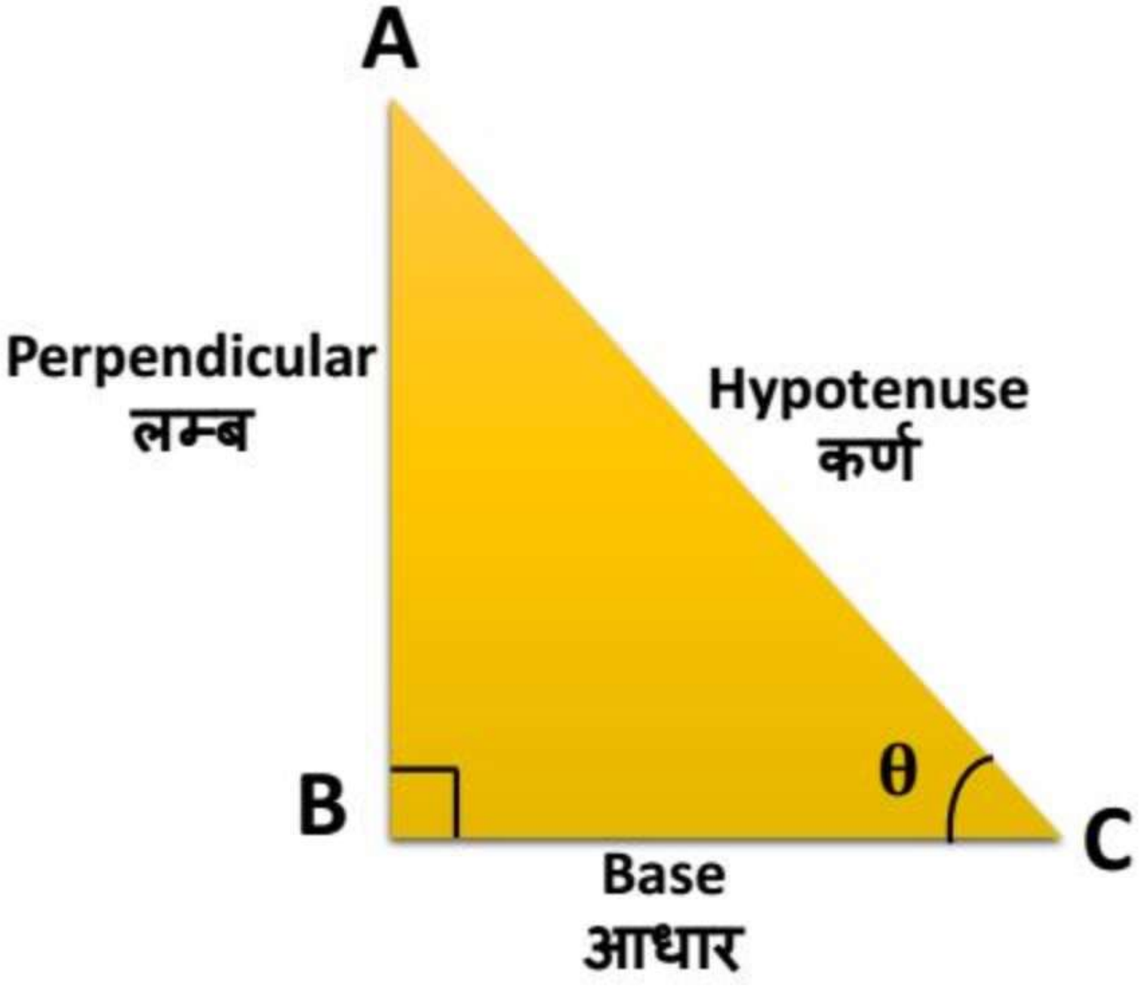
कितने त्रिकोणमितीय अनुपात होते हैं?

B → Base/आधार, P → Perpendicular/ लम्ब H → Hypotenuse/ कर्ण

(a) $\sin\theta = \frac{P}{H}$ (d) $\operatorname{cosec}\theta = \frac{H}{P}$

(b) $\cos\theta = \frac{B}{H}$ (e) $\sec\theta = \frac{H}{B}$

(c) $\tan\theta = \frac{P}{B}$ (f) $\cot\theta = \frac{B}{P}$



Note: It should be noted that (यह ध्यान दिया जाना चाहिए कि):

sin θ is an abbreviation for "sine of angle θ", it is not the product of sin and θ.

sin θ "कोण θ के ज्या" का संक्षिप्त नाम है, यह sin और θ का गुणनफल नहीं है।

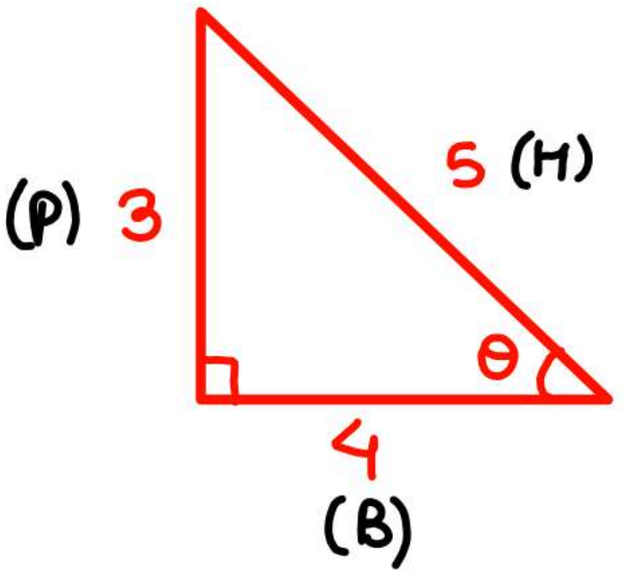
And $\boldsymbol{sin^2\theta = (sin\theta)^2, sin^3\theta = (sin\theta)^3, cos^3\theta = (cos\theta)^3, etc.}$

$$\text{cosec}\theta = \left(\frac{1}{\sin\theta}\right)$$

$$sec\theta = \left(\frac{1}{\cos\theta}\right)$$

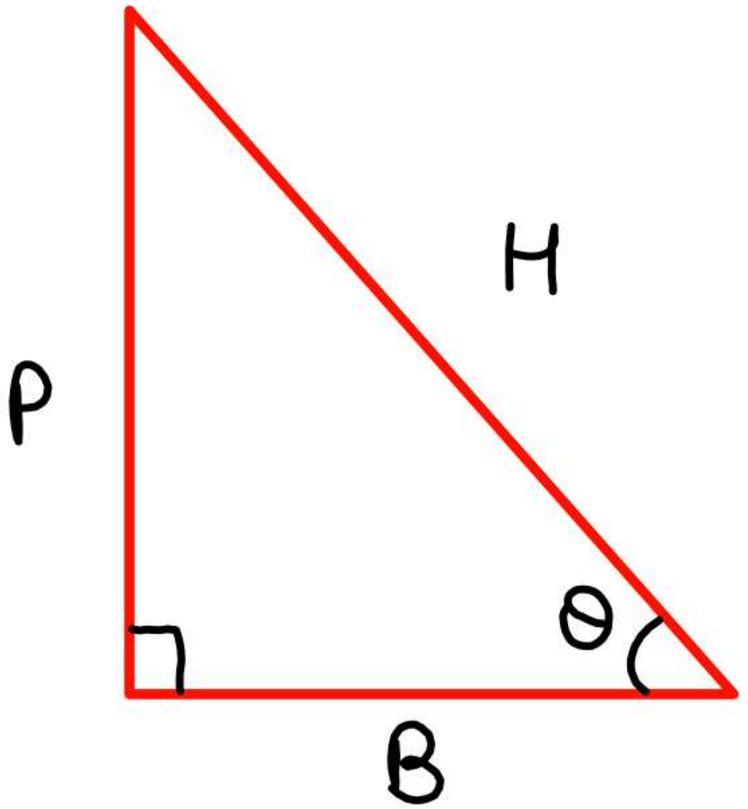
$$\cot\theta = \left(\frac{1}{\tan\theta}\right)$$

- $\sin\theta = \frac{P}{H} = \frac{3}{5}$
- $\cos\theta = \frac{B}{H} = \frac{4}{5}$
- $\tan\theta = \frac{P}{B} = \frac{3}{4}$
- $\operatorname{cosec}\theta = \frac{H}{P} = \frac{5}{3}$
- $\sec\theta = \frac{H}{B} = \frac{5}{4}$
- $\cot\theta = \frac{B}{P} = \frac{4}{3}$



PYTHAGORUS THEOREM

$$P^2 + B^2 = H^2$$



Pythagorean Triplets

$3^2+4^2=5^2$

$9+16 \quad 25$

$25=25$

$4^2+5^2=6^2$

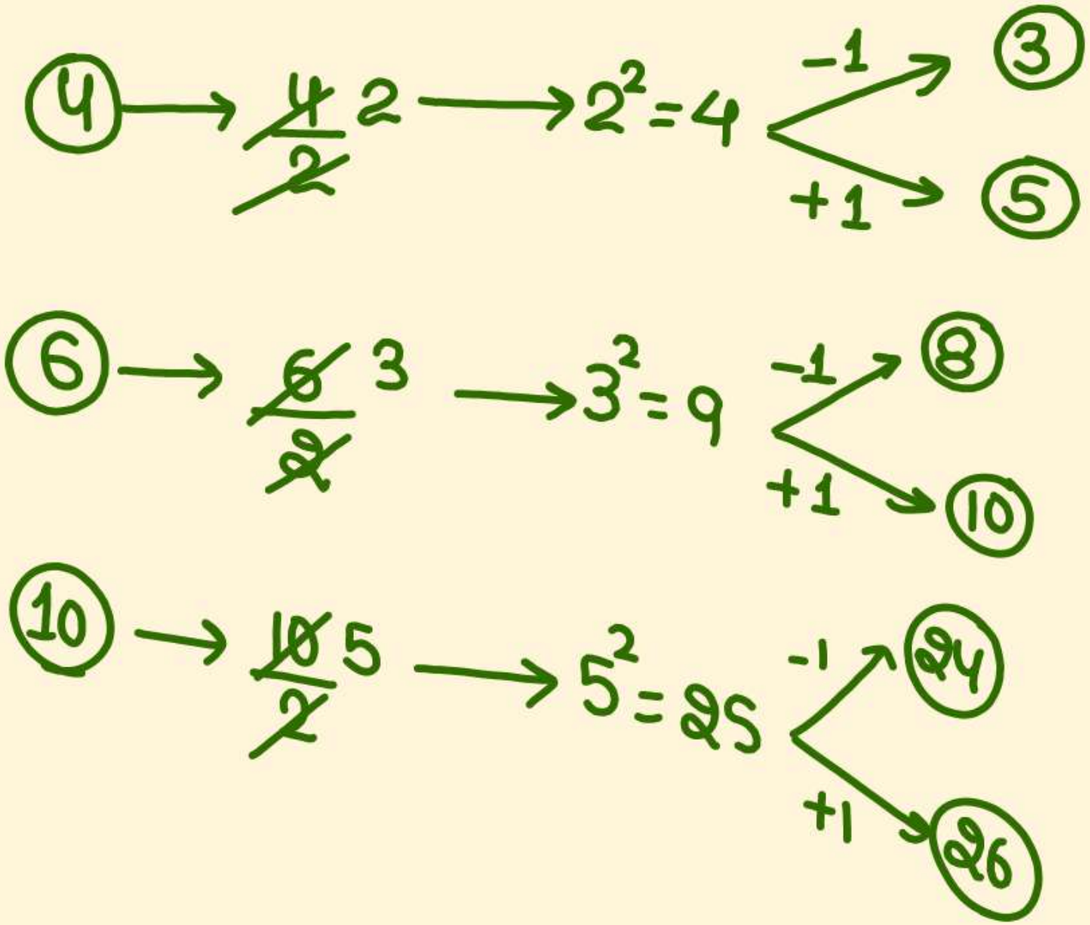
$16+25 \quad 36$

$41 \neq 36$

Special Triplets

· 3,4,5	(6,8,10) (9,12,15) (12,16,20)
· 5,12,13	$(5\sqrt{2},12\sqrt{2},13\sqrt{2})$ $(5\sqrt{3},12\sqrt{3},13\sqrt{3})$
· 6,8,10	(0.6,0.8,1.0) (3,4,5)
· 7,24,25	
· 8,15,17	
· 9,12,15	
· 9,40,41	
· 11,60,61	

How to make Triplets of Even Numbers



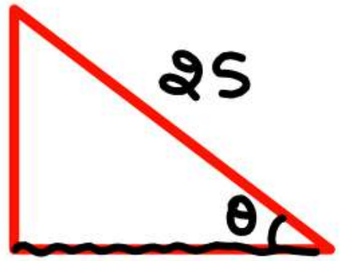
How to make Triplets of odd Numbers

- (3) → $3^2 = 9$ → $\frac{9}{2} = 4.5$ → (4) (-0.5), (5) (+0.5)
- (5) → $5^2 = 25$ → $\frac{25}{2} = 12.5$ → (12), (13)
- (7) → $7^2 = 49$ → $\frac{49}{2} = 24.5$ → (24), (25)

$\sin\theta = \frac{7}{25} \rightarrow \frac{P}{H}$

$\cos\theta = \frac{B}{H} = \frac{24}{25}$

$\tan\theta = \frac{P}{B} = \frac{7}{24}$



$\sqrt{25^2 - 7^2} = 24$

1. **If $\sin\theta = \frac{7}{25}$, then $\cos\theta$ and $\tan\theta$ are repsectively _____ .**
(where θ is in the first quadrant)

यदि $\sin\theta = \frac{7}{25}$ है, तो $\cos\theta$ और $\tan\theta$ के मान क्रमश: _____ हैं।
(जहाँ θ प्रथम चतुर्थांश में है)

SSC CHSL 09/07/2024 (Shift-03)

(a) $\frac{25}{24}$ and $\frac{24}{7}$

(b) $\frac{25}{7}$ and $\frac{7}{24}$

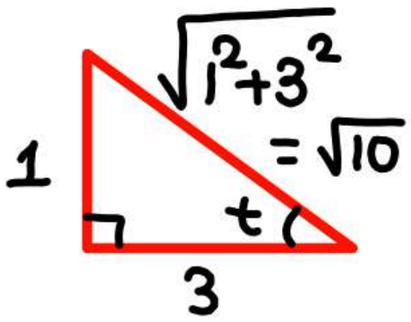
(c) $\frac{24}{25}$ and $\frac{24}{7}$

(d) ✓ $\frac{24}{25}$ and $\frac{7}{24}$

$\tan(t) = \frac{1}{3} \rightarrow \frac{P}{B}$

$\sec(t) = \frac{H}{B}$

$= \frac{\sqrt{10}}{3}$



2. What is the value of sec(t), if tan(t) = $\frac{1}{3}$?

यदि **tan(t)** = $\frac{1}{3}$ है, तो **sec(t)** का मान कितना है?

SSC CGL 09/09/2024 (Shift-02)

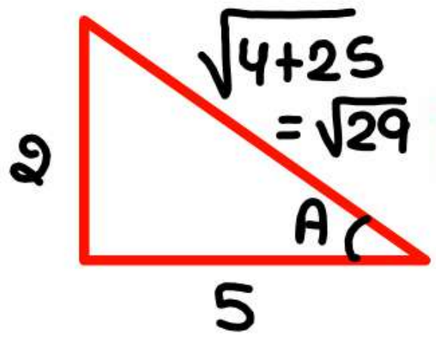
(a) $\frac{2\sqrt{2}}{3}$

(b) $\frac{\sqrt{10}}{9}$

(c) $\frac{\sqrt{10}}{3}$ ✓

(d) $\frac{\sqrt{3}}{3}$

$\tan A = \frac{2}{5}$ → P, 5 → B



$\frac{\sec^2 A}{\operatorname{cosec}^2 A} = \left(\frac{\frac{H^2}{B^2}}{\frac{H^2}{P^2}}\right)$

$= \frac{H^2 \times P^2}{H^2 \times B^2} = \frac{4}{25}$

3. If $\tan A = \frac{2}{5}$ find the value of $\frac{\sec^2 A}{\operatorname{cosec}^2 A}$.

यदि $\tan A = \frac{2}{5}$ है, तो $\frac{\sec^2 A}{\operatorname{cosec}^2 A}$ का मान ज्ञात कीजिए।

SSC CPO 03/10/2023 (Shift-3)

(a) $\frac{3}{5}$

(b) $\frac{4}{25}$

(c) $\frac{2}{5}$

(d) $\frac{9}{25}$

$$* \quad \frac{\sec\theta}{\operatorname{cosec}\theta} = \left(\frac{\frac{H}{B}}{\frac{H}{P}}\right) = \frac{H \times P}{H \times B} = \frac{P}{B} = \tan\theta$$

$$* \quad \frac{\operatorname{cosec}\theta}{\sec\theta} = \cot\theta$$

$$\left(\frac{\sec A}{\operatorname{cosec} A}\right)^2 = (\tan A)^2$$
$$= \frac{4}{25}$$

3. If $\tan A = \frac{2}{5}$ find the value of $\frac{\sec^2 A}{\operatorname{cosec}^2 A}$.

यदि $\tan A = \frac{2}{5}$ है, तो $\frac{\sec^2 A}{\operatorname{cosec}^2 A}$ का मान ज्ञात कीजिए।

SSC CPO 03/10/2023 (Shift-3)

(a) $\frac{3}{5}$

(b) $\frac{4}{25}$

(c) $\frac{2}{5}$

(d) $\frac{9}{25}$

4. If $\cot A = \frac{5}{12}$, then find the value of $\frac{\sqrt{\sec A - \operatorname{cosec} A}}{\sqrt{\sec A + \operatorname{cosec} A}} - \sqrt{\frac{17}{7}}$.

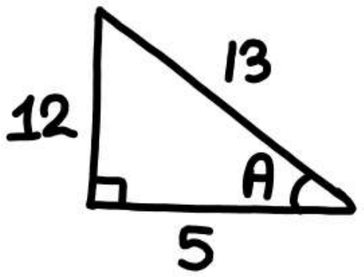
यदि $\cot A = \frac{5}{12}$ है, तो $\frac{\sqrt{\sec A - \operatorname{cosec} A}}{\sqrt{\sec A + \operatorname{cosec} A}} - \sqrt{\frac{17}{7}}$ का मान ज्ञात कीजिए।

SSC CHSL 08/07/2024 (Shift-01)

(a) $\frac{24}{\sqrt{119}}$ (b) 0

(c) $-\frac{10}{\sqrt{119}}$ (d) $\frac{10}{\sqrt{119}}$

$\cot A = \frac{5}{12} \rightarrow B$, $12 \rightarrow P$



$$\sqrt{\frac{\frac{13}{5} - \frac{13}{12}}{\frac{13}{5} + \frac{13}{12}}} - \sqrt{\frac{17}{7}}$$

$$= \sqrt{\frac{\frac{7}{60}}{\frac{17}{60}}} - \sqrt{\frac{17}{7}}$$

$$\left|\sqrt{\frac{7}{17}} - \sqrt{\frac{17}{7}}\right| = \frac{7-17}{\sqrt{119}} = \frac{-10}{\sqrt{119}}$$

5. If $\cos\theta = \frac{\sqrt{3}}{2}$, then $\tan^2\theta \cos^2\theta = ?$

यदि $\cos\theta = \frac{\sqrt{3}}{2}$ है, तो $\tan^2\theta \cos^2\theta = ?$

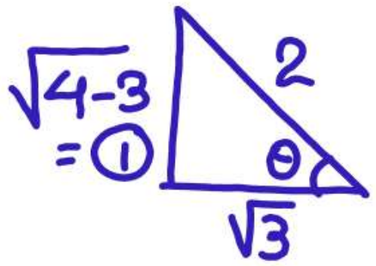
SSC CGL 14/07/2023 (Shift-1)

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) $\sqrt{3}$

$\cos\theta = \frac{\sqrt{3}}{2} \rightarrow B$

$2 \rightarrow H$



$\tan^2\theta \times \cos^2\theta$

$= \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$

$\tan\theta = \frac{7 \to P}{4 \to B}$

$$\frac{4\times\frac{P}{H} - 3\times\frac{B}{H}}{4\times\frac{P}{H} + 3\times\frac{B}{H}}$$

$$= \frac{4\times7 - 3\times4}{4\times7 + 3\times4} = \frac{16}{40}\ \frac{2}{5}$$

6. If $\tan\theta = \frac{7}{4}$, then the value of $\frac{4\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta}$ is equal to:

यदि $\tan\theta = \frac{7}{4}$ है, तो $\frac{4\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta}$ का मान ज्ञात कीजिए।

SSC CHSL 09/07/2024 (Shift-02)

(a) $\frac{2}{5}$ ✓

(b) $\frac{1}{3}$

(c) $\frac{4}{5}$

(d) $\frac{3}{5}$

$$\frac{4\tan\theta-3}{4\tan\theta+3}=\frac{4\times\frac{7}{4}-3}{4\times\frac{7}{4}+3}=\frac{2}{10}=\frac{1}{5}$$

6. If $\tan\theta=\frac{7}{4}$, then the value of $\frac{4\sin\theta-3\cos\theta}{4\sin\theta+3\cos\theta}$ is equal to:

यदि $\tan\theta=\frac{7}{4}$ है, तो $\frac{4\sin\theta-3\cos\theta}{4\sin\theta+3\cos\theta}$ का मान ज्ञात कीजिए।

SSC CHSL 09/07/2024 (Shift-02)

(a) $\frac{2}{5}$

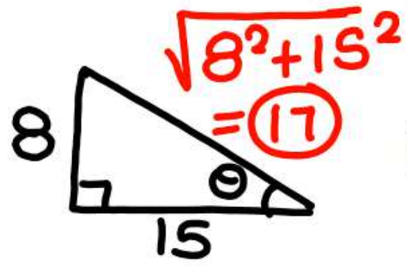
(b) $\frac{1}{3}$

(c) $\frac{4}{5}$

(d) $\frac{3}{5}$

$$\frac{\sin\theta}{\cos\theta}=\tan\theta$$

$\tan\theta = \frac{8}{15} \rightarrow \frac{P}{B}$



$$= \sqrt{\frac{1-\sin\theta}{1+\sin\theta}}$$

$$= \sqrt{\frac{1-\frac{8}{17}}{1+\frac{8}{17}}} = \sqrt{\frac{\frac{9}{17}}{\frac{25}{17}}} = \frac{3}{5}$$

7. **If $\tan\theta = \frac{8}{15}$, and θ is an acute angle, then the value of $\frac{\sqrt{1-\sin\theta}}{\sqrt{1+\sin\theta}}$ is:**

यदि $\tan\theta = \frac{8}{15}$ और θ कोई न्यूनकोण हो, तो $\frac{\sqrt{1-\sin\theta}}{\sqrt{1+\sin\theta}}$ का मान ज्ञात कीजिए।

SSC CGL 18/09/2024 (Shift-02)

(a) $\frac{2}{5}$

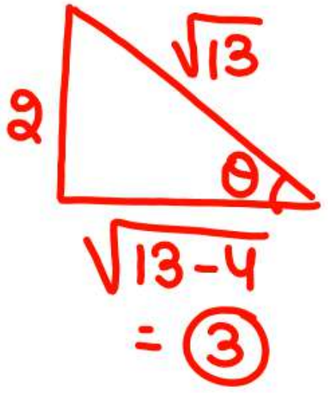
(b) $\frac{3}{5}$

(c) $\frac{1}{5}$

(d) $\frac{4}{5}$

$\sin\theta = \frac{2}{\sqrt{13}}$ → P, → H

$$\frac{\frac{3\times2}{3} + \frac{\sqrt{13}\times2}{\sqrt{13}}}{\frac{\sqrt{13}\times3}{\sqrt{13}} - \frac{3\times2}{3}} = \frac{4}{1}$$



8. If $\sqrt{13}\ \sin\theta = 2$, then the value of

$$\frac{3\tan\theta + \sqrt{13}\sin\theta}{\sqrt{13}\cos\theta - 3\tan\theta}$$ is:

यदि $\sqrt{13}\ \sin\theta = 2$ है, तो

$$\frac{3\tan\theta + \sqrt{13}\sin\theta}{\sqrt{13}\cos\theta - 3\tan\theta}$$ का मान ज्ञात कीजिए।

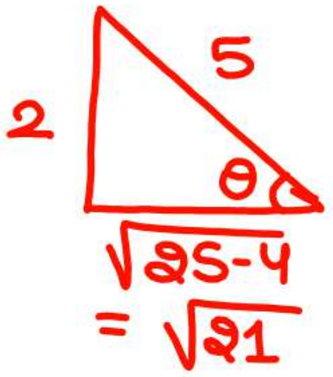
SSC CHSL 02/07/2024 (Shift-04)

Sinθ cosθ tanθ
P/H B/H P/B

(a) 4 (b) 3
(c) 2 (d) 0

$\sin A = \frac{2}{5}$ → P, 5 → H

$= \frac{5 \times \frac{2}{5} + 2 \times \frac{5}{2}}{\sqrt{21} \times \frac{5}{\sqrt{21}}} = \frac{7}{5}$



sinθ cosθ tanθ

P/H B/H P/B

9. **If $\sin A = \frac{2}{5}$, where A is an acute angle, what is the value of $\frac{5\sin A + 2\,\mathrm{cosec}\, A}{\sqrt{21}\sec A}$?**

यदि $\sin A = \frac{2}{5}$ है, जहाँ A एक न्यूनकोण है, तो $\frac{5\sin A + 2\,\mathrm{cosec}\, A}{\sqrt{21}\sec A}$ का मान क्या है?

SSC CHSL 10/08/2023 (Shift-01)

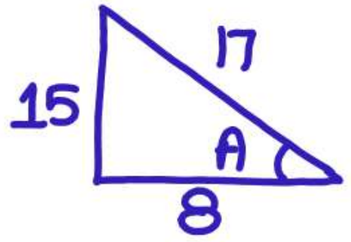
(a) $\frac{5}{4}$

(b) $\frac{7}{5}$

(c) $\frac{4}{5}$

(d) $\frac{5}{7}$

$\sec A = \frac{17}{8} \rightarrow H$, $8 \rightarrow B$



10. If $\sec A = \frac{17}{8}$, given that $A < 90°$, what is the value of the following? $\frac{34\sin A + 15\cot A}{68\cos A - 16\tan A}$

$\frac{\cancel{34}^{2} \times \frac{15}{\cancel{17}} + \cancel{15} \times \frac{8}{15}}{\cancel{4}\,\cancel{68} \times \frac{8}{\cancel{17}} - \cancel{16}^{4} \times \frac{15}{\cancel{8}}} = \frac{\cancel{38}}{\cancel{2}}$ (19)

यदि $\sec A = \frac{17}{8}$ है तो $A < 90°$ का मान क्या है? दिया है $\frac{34\sin A + 15\cot A}{68\cos A - 16\tan A}$

SSC CGL 11/04/2022 (Shift- 03)

(a) 23
(b) 19
(c) 30
(d) 38

$$\frac{7\tan\theta+5}{7\tan\theta-2} = \frac{3+5}{3-2} = \frac{8}{1}$$

11. If $7\tan\theta = 3$, then $\dfrac{7\sin\theta+5\cos\theta}{7\sin\theta-2\cos\theta}$ is:

यदि $7\tan\theta = 3$ है, तो $\dfrac{7\sin\theta+5\cos\theta}{7\sin\theta-2\cos\theta}$ का मान ज्ञात कीजिए।

SSC CGL 12/09/2024 (Shift-02)

(a) 7
(b) 5
(c) 8 ✓
(d) 6

12. H.W

Given that $\sin\theta = \frac{15}{17}$.

Evaluate $\frac{15\cot\theta + 17\sin\theta}{8\tan\theta + 16\sec\theta}$, where $0° < \theta < 90°$.

$\sin\theta = \frac{15}{17}$ दिया गया है। $\frac{15\cot\theta + 17\sin\theta}{8\tan\theta + 16\sec\theta}$ का मान ज्ञात कीजिए, जहां $0° < \theta < 90°$ है।

SSC CHSL 02/07/2024 (Shift-02)

(a) $\frac{49}{26}$

(b) $\frac{23}{49}$

(c) $\frac{49}{23}$

(d) $\frac{26}{23}$

13. If $\cos\theta = \frac{12}{13}$, then the value of $\frac{\sin\theta(1-\tan\theta)}{\tan\theta(1+\operatorname{cosec}\theta)}$ is :

H.W

यदि $\cos\theta = \frac{12}{13}$ है तो $\frac{\sin\theta(1-\tan\theta)}{\tan\theta(1+\operatorname{cosec}\theta)}$ का मान ज्ञात कीजिए।

SSC CGL MAINS 03/02/2022

(a) $\frac{25}{78}$

(b) $\frac{35}{234}$

(c) $\frac{35}{108}$

(d) $\frac{25}{156}$

$3\sin\theta - \cos\theta = \cos\theta + \sin\theta$

$\Rightarrow 2\sin\theta = 2\cos\theta$

$\Rightarrow 1 = \frac{\cos\theta}{\sin\theta}$

$1 = \cot\theta$

14. **If $\left\{\frac{(3\sin\theta - \cos\theta)}{(\cos\theta + \sin\theta)}\right\} = 1$, then the value of $\cot\theta$ is:**

यदि $\left\{\frac{(3\sin\theta - \cos\theta)}{(\cos\theta + \sin\theta)}\right\} = 1$ है, तो $\cot\theta$ का मान ज्ञात कीजिए।

SSC CGL 14/07/2023 (Shift-1)

(a) 3
(b) 0
(c) 1
(d) 2

15. If $\frac{7\sin\theta+4\cos\theta}{9\sin\theta-2\cos\theta}=\frac{5}{4}$, then the value of $\frac{\tan^2\theta+5}{\tan^2\theta-5}$ is:

यदि $\frac{7\sin\theta+4\cos\theta}{9\sin\theta-2\cos\theta}=\frac{5}{4}$ है, तो $\frac{\tan^2\theta+5}{\tan^2\theta-5}$ किसके बराबर होगा?

SSC CGL 09/09/2024 (Shift-03)

(a) $\frac{2121}{769}$

(b) $-\frac{2121}{769}$

(c) $\frac{769}{2121}$

(d) $-\frac{769}{2121}$

$28\sin\theta+16\cos\theta=45\sin\theta-10\cos\theta$

$\Rightarrow 26\cos\theta=17\sin\theta$

$\Rightarrow \frac{26}{17}=\tan\theta$

$\therefore \text{ans}=\frac{\frac{676}{289}+5}{\frac{676}{289}-5}=\frac{676+1445}{676-1445}$

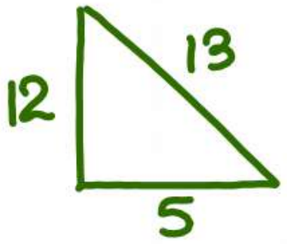
$=-\frac{2121}{769}$

$$\left(\frac{\frac{P^2}{H^2}-\frac{B^2}{H^2}}{2\times\frac{P}{H}\times\frac{B}{H}}\right)\times\frac{B^2}{P^2}$$

$$=\frac{(144-25)}{2\times12\times5}\times\frac{25}{144}\times5$$

$$=\frac{595}{3456}$$

16. If $\sin\theta = \frac{12}{13}$ ($12 \to P$, $13 \to H$), $0° < \theta < 90°$, then



$$\frac{\sin^2\theta-\cos^2\theta}{2\sin\theta.\cos\theta}\times\frac{1}{\tan^2\theta}=$$ _____.

यदि $\sin\theta = \frac{12}{13}$, $0° < \theta < 90°$ है तो

$$\frac{\sin^2\theta-\cos^2\theta}{2\sin\theta.\cos\theta}\times\frac{1}{\tan^2\theta}=$$ _____.

SSC CHSL 08/06/2022 (Shift- 2)

(a) $\frac{295}{3456}$

(b) $\frac{290}{3542}$

(c) $\frac{695}{3542}$

(d) ✓ $\frac{595}{3456}$

17. If $\tan\theta = \frac{2}{\sqrt{11}}$, $0° < \theta < 90°$, then the value of $\frac{2\text{cosec}^2\theta - 3\sec^2\theta}{3\text{cosec}^2\theta + 4\sec^2\theta}$ is equal to :

H.W

यदि $\tan\theta = \frac{2}{\sqrt{11}}$, $0° < \theta < 90°$ है तो $\frac{2\text{cosec}^2\theta - 3\sec^2\theta}{3\text{cosec}^2\theta + 4\sec^2\theta}$ का मान ज्ञात कीजिए।

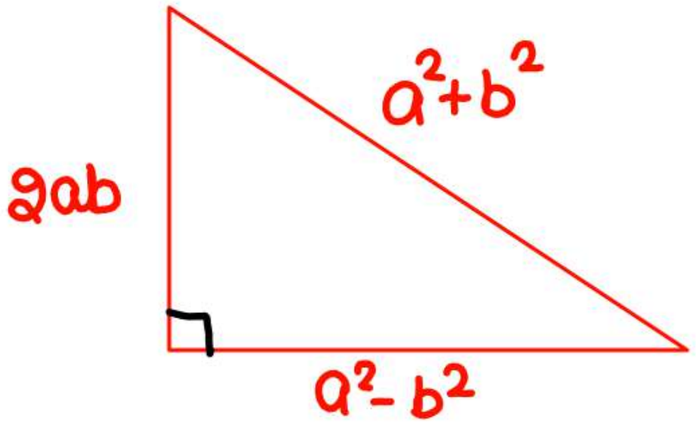
CHSL 26/10/2020 (Shift- 03)

(a) $\frac{11}{45}$

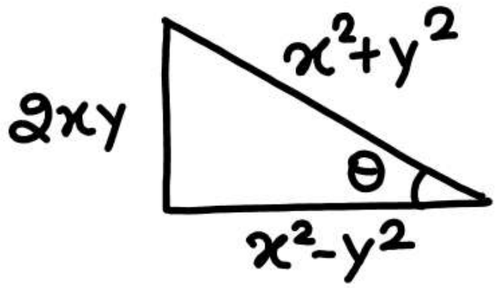
(b) $\frac{11}{49}$

(c) $\frac{13}{49}$

(d) $\frac{10}{49}$



	P	B	H
	$2ab$	a^2-b^2	a^2+b^2
$a=2, b=1$	4	3	5
$a=3, b=2$	12	5	13



$$\tan\theta = \frac{2xy}{x^2-y^2}$$

18. If $\sin\theta = \dfrac{2xy}{x^2+y^2}$, then the value of $\tan\theta$ is:

यदि $\sin\theta = \dfrac{2xy}{x^2+y^2}$ तो $\tan\theta$ का मान है:

SSC CHSL 01/07/2024 (Shift-02)

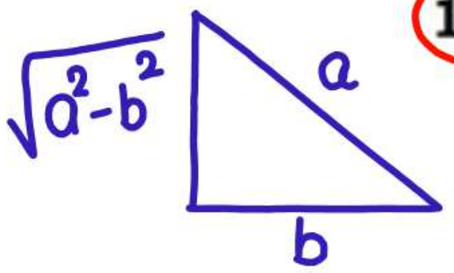
(a) $\dfrac{2xy}{x^2-y^2}$

(b) $\dfrac{x^2+y^2}{x^2-y^2}$

(c) $\dfrac{x^2-y^2}{2xy}$

(d) $\dfrac{x^2-y^2}{x^2+y^2}$

$$\frac{1-\left(\frac{a^2-b^2}{b^2}\right)}{2-\frac{(a^2-b^2)}{a^2}}$$



$$= \frac{\frac{b^2-(a^2-b^2)}{b^2}}{\frac{2a^2-(a^2-b^2)}{a^2}} = \left(\frac{\frac{b^2-a^2+b^2}{b^2}}{\frac{2a^2-a^2+b^2}{a^2}}\right)$$

$$= \frac{a^2(2b^2-a^2)}{b^2(a^2+b^2)}$$

19. If $\sec\theta = \frac{a}{b}$, $b \neq 0$, then $\frac{1-\tan^2\theta}{2-\sin^2\theta} = ?$

यदि $\sec\theta = \frac{a}{b}$ ($a \to H$, $b \to B$), $b \neq 0$ है तो $\frac{1-\tan^2\theta}{2-\sin^2\theta} = ?$

SSC CGL MAINS (15/10/2020)

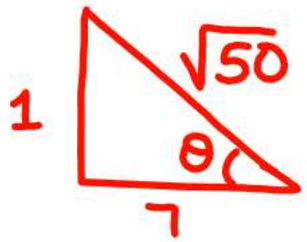
(a) $\frac{a^2(2b^2+a^2)}{b^2(a^2-b^2)}$

(b) $\frac{a^2(2b^2+a^2)}{b^2(a^2+b^2)}$

(c) $\frac{a^2(2b^2-a^2)}{b^2(a^2+b^2)}$ ✓

(d) $\frac{a^2(2b^2-a^2)}{a^2(a^2+b^2)}$

$\cot A = \frac{7}{1} \rightarrow \frac{B}{P}$



20. If cot A = 7, then the value of $\frac{5\cos A + 4\sin A}{\cos^3 A + 7\sin^3 A + 6\sin A}$ is:

यदि cot A = 7 है, तो $\frac{5\cos A + 4\sin A}{\cos^3 A + 7\sin^3 A + 6\sin A}$ का मान ज्ञात करें।

SSC CHSL 01/07/2024 (Shift-03)

(a) 3
(b) 2
(c) 0
(d) 1

$$\frac{5\times\frac{7}{\sqrt{50}} + 4\times\frac{1}{\sqrt{50}}}{\frac{343}{50\sqrt{50}} + 7\times\frac{1}{50\sqrt{50}} + 6\times\frac{1}{\sqrt{50}}}$$

$$= \frac{39}{\frac{343+7+300}{50}} = \frac{39\times 50}{650} = 3$$

$(\sqrt{3})^3 = 3\sqrt{3}$

$(\sqrt{5})^3 = \sqrt{5}\times\sqrt{5}\times\sqrt{5}$

$= 5\sqrt{5}$

21. If $\frac{\text{cosec}\,\theta+\cot\theta}{\text{cosec}\,\theta-\cot\theta}=7$, then the value of $\frac{4\sin^2\theta-1}{4\sin^2\theta+5}$ is:

यदि $\frac{\text{cosec}\,\theta+\cot\theta}{\text{cosec}\,\theta-\cot\theta}=7$ है तो $\frac{4\sin^2\theta-1}{4\sin^2\theta+5}$ का मान ज्ञात कीजिए।

SSC CGL 16/08/2021 (Shift- 01)

(a) $\frac{1}{3}$

(b) $-\frac{1}{3}$

(c) $-\frac{1}{9}$

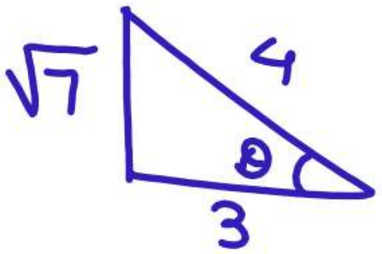
(d) $\frac{1}{9}$

$\text{cosec}\theta+\cot\theta=7\,\text{cosec}\theta-7\cot\theta$

$8\cot\theta=6\,\text{cosec}\theta$

$\Rightarrow 4\times\frac{B}{P}=3\times\frac{H}{P}$

$\Rightarrow \frac{B}{H}=\frac{3}{4}$



$$\frac{4\times\frac{7}{16}-1}{4\times\frac{7}{16}+5}=\frac{4\times4-16}{\frac{4\times7+80}{16}}$$

$$=\frac{12}{108}=\frac{1}{9}$$

Componendo & Dividendo

$$\frac{a+b}{a-b} = \frac{11}{5}$$

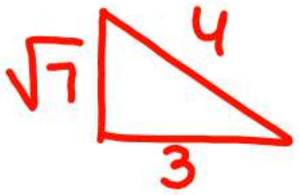
$$\frac{a}{b} = \frac{11+5}{11-5} = \frac{16}{6} \quad \frac{8}{3}$$

$$\frac{a+b}{a-b} = \frac{5}{3}$$

$$\frac{a}{b} = \frac{8}{2} \quad \frac{4}{1}$$

$\frac{\text{cosec}\theta}{\cot\theta} = \frac{5}{3}$

$\sec\theta = \frac{4}{3}$



$\frac{4\times\frac{7}{16}-1}{4\times\frac{7}{16}+5} = \frac{28-16}{28+80} = \frac{12}{108} \quad \frac{1}{9}$

21. If $\frac{\text{cosec}\,\theta+\cot\theta}{\text{cosec}\,\theta-\cot\theta}=7$, then the value of $\frac{4\sin^2\theta-1}{4\sin^2\theta+5}$ is:

यदि $\frac{\text{cosec}\,\theta+\cot\theta}{\text{cosec}\,\theta-\cot\theta}=7$ है तो $\frac{4\sin^2\theta-1}{4\sin^2\theta+5}$ का मान ज्ञात कीजिए।

SSC CGL 16/08/2021 (Shift- 01)

(a) $\frac{1}{3}$

(b) $-\frac{1}{3}$

(c) $-\frac{1}{9}$

(d) $\frac{1}{9}$ ✓

$\frac{\sec\theta}{\tan\theta} = \text{cosec}\,\theta \qquad \frac{\text{cosec}\,\theta}{\cot\theta} = \sec\theta$

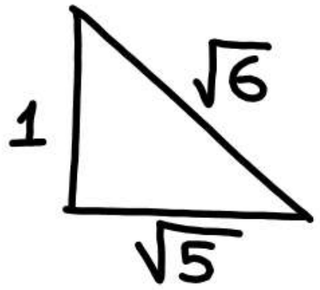
$S^2 = 5T^2 - 5S^2$

$\Rightarrow 6S^2 = 5S^2$

C^2

$C^2 = \frac{5}{6}$

$\therefore C = \frac{\sqrt{5}}{\sqrt{6}}$



$\frac{24 \times \frac{4}{6} - 15 \times \frac{3}{5}}{6 \times \frac{6}{1} - 7 \times \frac{5}{1}} = \frac{-14}{1}$

22. If $\frac{\sin^2\theta}{\tan^2\theta - \sin^2\theta} = 5$, θ is an acute angle, then the value of $\frac{24\sin^2\theta - 15\sec^2\theta}{6\operatorname{cosec}^2\theta - 7\cot^2\theta}$ is:

यदि $\frac{\sin^2\theta}{\tan^2\theta - \sin^2\theta} = 5$, θ न्यूनकोण है तो

$\frac{24\sin^2\theta - 15\sec^2\theta}{6\operatorname{cosec}^2\theta - 7\cot^2\theta}$ का मान ज्ञात कीजिए।

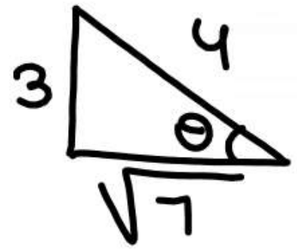
SSC CGL 23/08/2021 (Shift- 02)

(a) 2
(b) –14
(c) 14
(d) –2

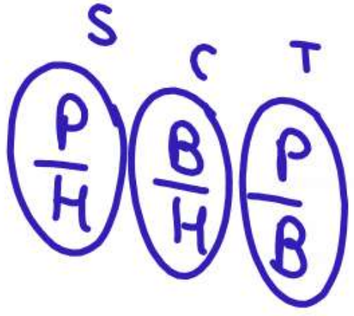
$$\frac{\sec\theta+\tan\theta}{\sec\theta-\tan\theta}=\frac{7}{1}$$

$$\frac{\sec\theta}{\tan\theta}=\frac{8}{6}\,\frac{4}{3}$$

$$\operatorname{cosec}\theta=\frac{4}{3}$$



$$\frac{\frac{4}{3}+\frac{7}{9}}{\frac{4}{3}-\frac{7}{9}}=\frac{\frac{19}{9}}{\frac{5}{9}}=\frac{19}{5}$$



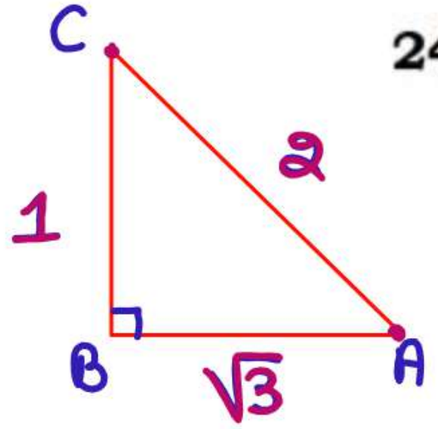
23. If $\frac{\sec\theta-\tan\theta}{\sec\theta+\tan\theta}=\frac{1}{7}$, θ lies in first quadrant, then the value of $\frac{\operatorname{cosec}\theta+\cot^2\theta}{\operatorname{cosec}\theta-\cot^2\theta}$ is:

यदि $\frac{\sec\theta-\tan\theta}{\sec\theta+\tan\theta}=\frac{1}{7}$, θ प्रथम चतुर्थांश में स्थित है तो $\frac{\operatorname{cosec}\theta+\cot^2\theta}{\operatorname{cosec}\theta-\cot^2\theta}$ का मान है।

SSC CGL MAINS 03/02/2022

(a) $\frac{19}{5}$ (b) $\frac{22}{3}$

(c) $\frac{37}{12}$ (d) $\frac{37}{19}$



24. In a ΔABC, right angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, then $\sin A.\cos C + \cos A.\sin C$ = _____.

एक ΔABC में, B पर समकोण है, यदि $\tan A = \frac{1}{\sqrt{3}}$ है, तो $\sin A.\cos C + \cos A.\sin C$ का मान ज्ञात कीजिए।

SSC CHSL 26/05/2022 (Shift- 1)

(a) 0
(b) 2
(c) – 1
(d) 1

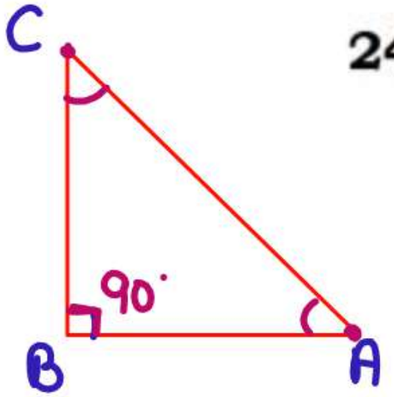
$\sin A \times \cos C + \cos A \times \sin C$

$= \frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}$

$= \frac{1}{4} + \frac{3}{4}$

$= \frac{4}{4} = 1$

$$\sin(A+C) = \sin A \cos C + \cos A \sin C$$



24. In a ΔABC, right angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, then $\sin A \cdot \cos C + \cos A \cdot \sin C =$ _____.

एक ΔABC में, B पर समकोण है, यदि $\tan A = \frac{1}{\sqrt{3}}$ है, तो $\sin A \cdot \cos C + \cos A \cdot \sin C$ का मान ज्ञात कीजिए।

SSC CHSL 26/05/2022 (Shift- 1)

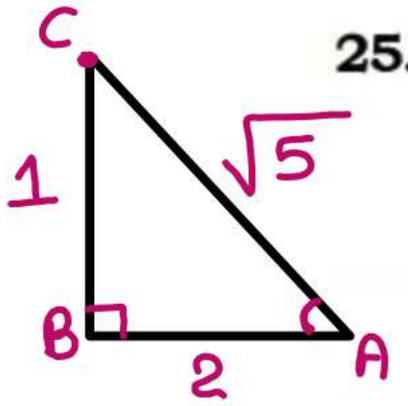
(a) 0
(b) 2
(c) – 1
(d) 1

$\sin A \times \cos C + \cos A \times \sin C$

$= \sin(A+C)$

$= \sin 90°$

$= 1$



25. In ΔABC, right angled at B, if $\tan A = \frac{1}{2}$, then the value of $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$ is :

एक ΔABC में, B पर समकोण है, यदि $\tan A = \frac{1}{2}$ है, तो $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$ का मान ज्ञात कीजिए।

SSC CGL TIER-II (16/10/2020)

(a) $2\sqrt{5}$ (b) 3
(c) 2 (d) 1

A: P = 1, B = 2, H = $\sqrt{5}$

$\sin A = \frac{1}{\sqrt{5}}$, $\cos A = \frac{2}{\sqrt{5}}$

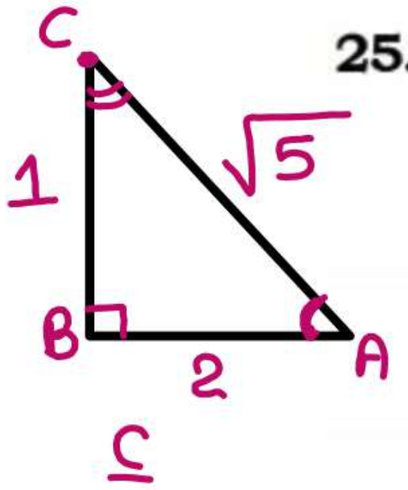
C: P = 2, B = 1, H = $\sqrt{5}$

$\sin C = \frac{2}{\sqrt{5}}$, $\cos C = \frac{1}{\sqrt{5}}$

$$\frac{\sqrt{5}\left(\frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}}\right)}{\sqrt{5}\left(\frac{2}{\sqrt{5}} - \frac{1}{\sqrt{5}}\right)} = \frac{3}{1}$$

$$\frac{\frac{1}{\sqrt{5}}\left(\frac{1}{\sqrt{5}}+\frac{2}{\sqrt{5}}\right)}{\frac{1}{\sqrt{5}}\left(\frac{2}{\sqrt{5}}-\frac{1}{\sqrt{5}}\right)}$$

$$= \frac{\frac{3}{\sqrt{5}}}{\frac{1}{\sqrt{5}}} = 3$$

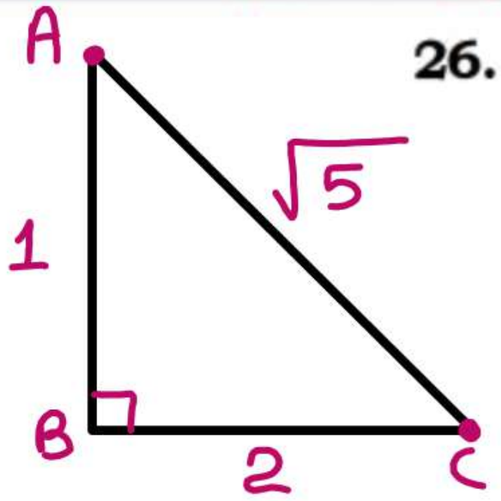


25. In ΔABC, right angled at B, if $\tan A = \frac{1}{2}$, then the value of $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$ is :

एक ΔABC में, B पर समकोण है, यदि $\tan A = \frac{1}{2}$ है, तो $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$ का मान ज्ञात कीजिए।

SSC CGL TIER-II (16/10/2020)

(a) $2\sqrt{5}$
(b) 3
(c) 2
(d) 1



26. In ΔABC, $\angle B = 90°$ and AB : BC = 1 : 2. The value of cos A + tan C is:

ΔABC में, $\angle B = 90°$ और AB : BC = 1 : 2 है। cos A + tan C का मान_____ है।

SSC CPO 04/10/2023 (Shift-01)

(a) $\frac{1+\sqrt{5}}{2\sqrt{5}}$

(b) $\frac{2\sqrt{5}}{2+\sqrt{5}}$

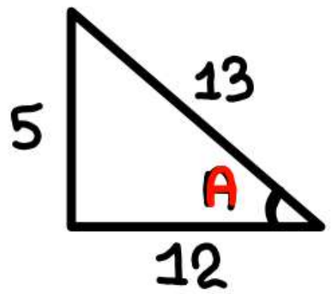
(c) $\frac{5+\sqrt{5}}{2\sqrt{5}}$

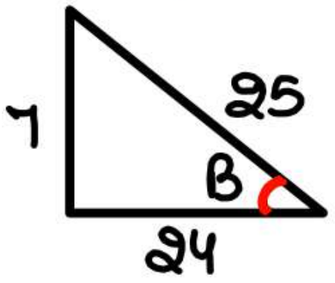
(d) $\frac{2+\sqrt{5}}{2\sqrt{5}}$

$= \frac{1}{\sqrt{5}} + \frac{1}{2}$

$= \frac{2+\sqrt{5}}{2\sqrt{5}}$

$\sin A = \frac{5 \to P}{13 \to H}$ $\cot B = \frac{24 \to B}{7 \to P}$





$= \frac{13}{\cancel{12}} \times \frac{\cancel{24}^{2}}{\cancel{25}} \times \frac{\cancel{25}}{7} \times \frac{5}{\cancel{12}_{6}} = \frac{65}{42}$

27. If $\sin A = \frac{5}{13}$ and $7\cot B = 24$, then the value of (secA cosB)(cosecBtan A) is:

यदि $\sin A = \frac{5}{13}$ तथा $7\cot B = 24$ है तो (secA cosB)(cosecBtan A) का मान ज्ञात कीजिए।

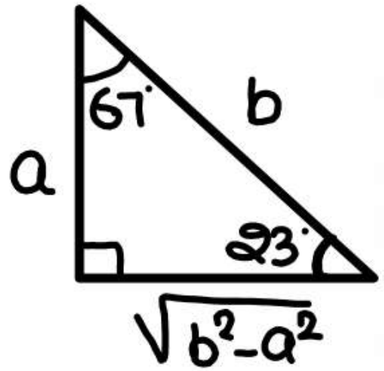
SSC CGL MAINS/29/01/2022

(a) $\frac{13}{7}$

(b) $\frac{65}{42}$

(c) $\frac{15}{13}$

(d) $\frac{13}{14}$



$$\frac{b}{\sqrt{b^2-a^2}} - \frac{\sqrt{b^2-a^2}}{b}$$

$$= \frac{b^2-(b^2-a^2)}{b\sqrt{b^2-a^2}} = \frac{b^2-b^2+a^2}{b\sqrt{b^2-a^2}}$$

28. **If $\sin 23^\circ = \frac{a}{b}$, then the value of $\sec 23^\circ - \sin 67^\circ$ is _____.**

यदि $\sin 23^\circ = \frac{a}{b}$ ($\frac{P}{H}$) है, तो $\sec 23^\circ - \sin 67^\circ$ का मान क्या है?

SSC CGL 14/07/2023 (Shift-3)

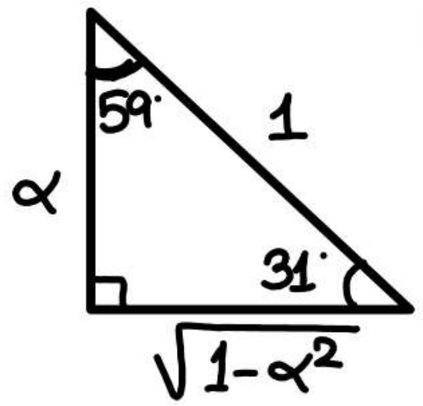
(a) $\frac{a^2}{\sqrt{b^2-a^2}}$

(b) $\frac{b^2-a^2}{ab}$

(c) $\frac{a^2}{b\sqrt{b^2+a^2}}$

(d) $\frac{a^2}{b\sqrt{b^2-a^2}}$

$\sin 31 = \frac{\alpha}{1} \rightarrow \frac{P}{H}$



$\cot 59° = \frac{B}{P} = \frac{\alpha}{\sqrt{1-\alpha^2}}$

29. **If $\sin 31° = \alpha$, then the value of $\cot 59°$ is:**

यदि $\sin 31° = \alpha$ है, तो $\cot 59°$ का मान ज्ञात कीजिए।

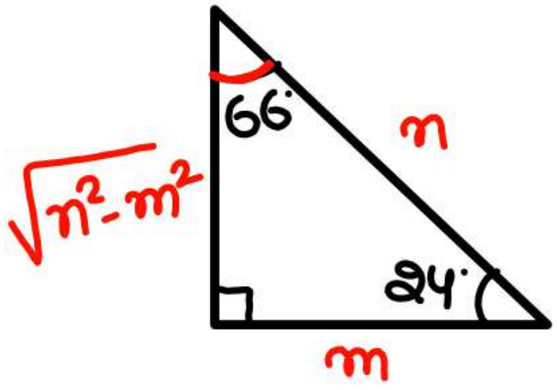
SSC CGL 13/09/2024 (Shift-01)

(a) $\frac{\sqrt{1-\alpha^2}}{2\alpha}$

(b) $\frac{\alpha}{\sqrt{1-\alpha^2}}$

(c) $\frac{\sqrt{1+\alpha^2}}{\alpha}$

(d) $\frac{\alpha}{\sqrt{1+\alpha^2}}$



30. If $\cos 24° = \frac{m}{n}$, then the value of (cosec24° – cos 66°) is:

यदि $\cos 24° = \frac{m}{n}$ है, तो (cosec24° – cos 66°) का मान क्या होगा?

SSC CGL 17/09/2024 (Shift-01)

(a) $\frac{m^2}{n\sqrt{m^2-n^2}}$

(b) $\frac{n^2}{n\sqrt{m^2-n^2}}$

(c) $\frac{n^2}{m\sqrt{n^2-m^2}}$

(d) $\frac{m^2}{n\sqrt{n^2-m^2}}$

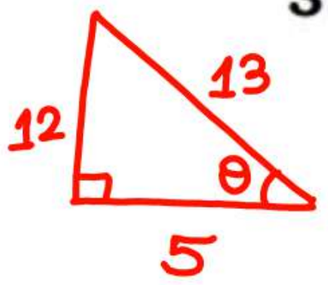
$\text{Cosec } 24° - \cos 66°$

$= \frac{n}{\sqrt{n^2-m^2}} - \frac{\sqrt{n^2-m^2}}{n}$

$= \frac{n^2-(n^2-m^2)}{n\sqrt{n^2-m^2}} = \frac{m^2}{n\sqrt{n^2-m^2}}$

$\frac{P}{H} - \frac{B}{H} = \frac{7}{13}$

$\Rightarrow \frac{P-B}{H} = \frac{7}{13}$



$\sin\theta + \cos\theta$

$= \frac{12}{13} + \frac{5}{13} = \frac{17}{13}$

31. If $\sin\theta - \cos\theta = \frac{7}{13}$, $0° < \theta < 90°$, then the value of $\sin\theta + \cos\theta$ is :

यदि $\sin\theta - \cos\theta = \frac{7}{13}$, $0° < \theta < 90°$ है तो $\sin\theta + \cos\theta$ का मान ज्ञात कीजिए।

(a) $\frac{17}{13}$

(b) $\frac{13}{17}$

(c) $\frac{1}{13}$

(d) $\frac{1}{17}$

32. If $0° < A, B < 45°$, $\cos(A+B) = \frac{24}{25}$ and $\sin(A-B) = \frac{15}{17}$, then $\tan 2A = ?$

Hold

यदि $0° < A, B < 45°$, $\cos(A+B) = \frac{24}{25}$ तथा $\sin(A-B) = \frac{15}{17}$ है तो $\tan 2A$ का मान ज्ञात कीजिए।

SSC CGL 06/03/2020 (Shift- 02)

(a) 0

(b) 1

(c) $\frac{416}{87}$

(d) $\frac{213}{4}$

QUESTIONS BASED ON VALUES OF TRIGONOMETRIC RATIOS

	0°	30°	45°	60°	90°
Sinθ	$\sqrt{\frac{0}{4}} = \frac{0}{1}$	$\sqrt{\frac{1}{4}} = \frac{1}{2}$	$\sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}}$	$\sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$	$\sqrt{\frac{4}{4}} = 1$
Cosθ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\frac{0}{1}$
tanθ	$\frac{0}{1} = \frac{0}{1}$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\frac{1}{0} = N.D$
Cosecθ	$\frac{1}{0} = N.D$	$\frac{2}{1}$	$\frac{\sqrt{2}}{1}$	$\frac{2}{\sqrt{3}}$	1
Secθ	1	$\frac{2}{\sqrt{3}}$	$\frac{\sqrt{2}}{1}$	$\frac{2}{1}$	$\frac{1}{0} = N.D$
Cotθ	$\frac{1}{0} = N.D$	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	$\frac{0}{1} = 0$

	0°	30°	45°	60°	90°
↑ Sinθ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
↓ Cosθ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\frac{0}{1}$
↑ tanθ	$\frac{0}{1}=\frac{0}{1}$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\frac{1}{0}=\text{N.D}$
Cosecθ	$\frac{1}{0}=\text{N.D}$	$\frac{2}{1}$	$\frac{\sqrt{2}}{1}$	$\frac{2}{\sqrt{3}}$	1
Secθ	1	$\frac{2}{\sqrt{3}}$	$\frac{\sqrt{2}}{1}$	$\frac{2}{1}$	$\frac{1}{0}=\text{N.D}$
Cotθ	$\frac{1}{0}=\text{N.D}$	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	$\frac{0}{1}=0$

Angle

- Degree: 1°
- Radian: 1^c

$$180^\circ = \pi^c$$

or

$$\pi^c = 180^\circ$$

$$\frac{22}{7}{}^c = 180^\circ$$

$$1^c = \frac{180 \times 7}{22} = 57.27^\circ$$

$$1^c = 57.27^\circ$$

For 0°-90°

① $\sin 10^\circ < \sin 20^\circ$

② $\sin 20^\circ < \sin 30^\circ$

③ $\sin 10^\circ < \sin 80^\circ$

① $\cos 10^\circ > \cos 20^\circ$

② $\cos 30^\circ > \cos 60^\circ$

③ $\cos 10^\circ > \cos 80^\circ$

① $\tan 10^\circ < \tan 20^\circ$

② $\tan 20^\circ < \tan 80^\circ$

③ $\tan 30^\circ < \tan 60^\circ$

$\sin 1^c$ $\quad$ $\sin 10^\circ$

$\sin 57^\circ > \sin 10^\circ$

33. Simplify $\dfrac{\sin 60° - \cos 60°}{\sin 60° + \cos 60°}$

$$\frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{\frac{\sqrt{3}}{2} + \frac{1}{2}} = \frac{\sqrt{3}-1}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)}$$

$$= \frac{(\sqrt{3}-1)^2}{(3-1)} = \frac{3+1-2\sqrt{3}}{2}$$

$$= \frac{4-2\sqrt{3}}{2} = (2-\sqrt{3})$$

$\dfrac{\sin 60° - \cos 60°}{\sin 60° + \cos 60°}$ को सरल कीजिए।

SSC CHSL 11/07/2024 (Shift-03)

(a) $2 + \sqrt{3}$ (b) $\sqrt{3} - 1$

(c) $2 - \sqrt{3}$ (d) $\sqrt{3} - 2$

$\sin 60° = \frac{\sqrt{3}}{2}$

$\cos 60° = \frac{1}{2}$

34. What is the value of sin30° + cos30° + tan30°

sin30° + cos30° + tan30° का मान क्या है?

SSC CGL 14/07/2023 (Shift-3)

(a) $\frac{5+\sqrt{3}}{2\sqrt{3}}$

(b) $\frac{5-\sqrt{3}}{2\sqrt{3}}$

(c) $\frac{5-\sqrt{3}}{\sqrt{3}}$

(d) $\frac{5-\sqrt{3}}{\sqrt{3}}$

$\frac{1}{2}+\frac{\sqrt{3}}{2}+\frac{1}{\sqrt{3}}$

$= \frac{\sqrt{3}+3+2}{2\sqrt{3}} = \frac{5+\sqrt{3}}{2\sqrt{3}}$

35. The value of $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$ at $\theta = 45°$ is:

$\theta = 45°$ पर $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$ का मान ज्ञात कीजिए।

SSC CGL 23/09/2024 (Shift-03)

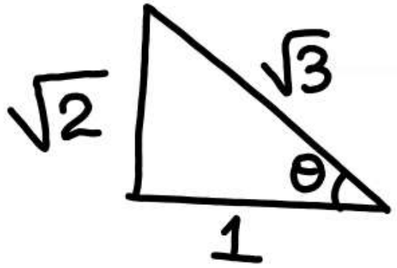
(a) 1
(b) 0
(c) 3
(d) 2

$$\frac{\left(\frac{1}{\sqrt{2}}\right)^2 - 2\times\left(\frac{1}{\sqrt{2}}\right)^4}{2\times\left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^2} = \frac{\frac{1}{2} - 2\times\frac{1}{4}}{2\times\frac{1}{4} + \frac{1}{2}}$$

$= 0$

$\sqrt{3} \times \frac{S}{C} = \sqrt{3} \times \sqrt{3} \times S$

$\cos\theta = \frac{1}{\sqrt{3}}$ → B, H



$\text{ans} = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$

36. **If $\sqrt{3}\tan\theta = 3\sin\theta$, then what is the value of $\sin^2\theta - \cos^2\theta$?**

यदि $\sqrt{3}\tan\theta = 3\sin\theta$ है तो $\sin^2\theta - \cos^2\theta$ का मान ज्ञात कीजिए।

SSC CGL MAINS 08/08/2022

(a) $\frac{1}{2}$ (b) $\frac{1}{5}$

(c) $\frac{1}{3}$ ✓ (d) $\frac{1}{4}$

$\sqrt{3} \times 5 = \sqrt{3} \times \sqrt{3} \times 5$

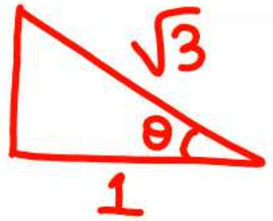
C

B

$\frac{1}{\sqrt{3}} = \cos\theta$

H

$\sqrt{2}$



$\sin^2\theta - \cos^2\theta$

$= \frac{2}{3} - \frac{1}{3}$

$= \frac{1}{3}$

36. **If $\sqrt{3}\tan\theta = 3\sin\theta$, then what is the value of $\sin^2\theta - \cos^2\theta$?**

यदि $\sqrt{3}\tan\theta = 3\sin\theta$ है तो $\sin^2\theta - \cos^2\theta$ का मान ज्ञात कीजिए।

SSC CGL MAINS 08/08/2022

(a) $\frac{1}{2}$

(b) $\frac{1}{5}$

(c) $\frac{1}{3}$

(d) $\frac{1}{4}$

$$2k \times \frac{1}{2} \times \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} = \frac{(\sqrt{3})^2 \times 2 \times 1}{(\sqrt{2})^2 \times 2}$$

$$\frac{k}{2} = \frac{3}{2}$$

37. If $2k\sin 30^\circ\cos 30^\circ\cot 60^\circ = \frac{\cot^2 30^\circ \sec 60^\circ \tan 45^\circ}{\text{cosec}^2 45^\circ \text{cosec} 30^\circ}$, then find the value of k.

यदि $2k\sin 30^\circ\cos 30^\circ\cot 60^\circ = \frac{\cot^2 30^\circ \sec 60^\circ \tan 45^\circ}{\text{cosec}^2 45^\circ \text{cosec} 30^\circ}$ है तो **k** का मान ज्ञात कीजिए।

SSC CGL 12/04/2022 (Shift- 03)

(a) $\frac{3}{2}$

(b) 3

(c) 1

(d) 6

$$\frac{12\sin 30^\circ + 5\cos 45^\circ}{9\sin 45^\circ - 4\cos 60^\circ}$$

$$= \frac{12\times\frac{1}{2} + 5\times\frac{1}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}}}{9\times\frac{1}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}} - 4\times\frac{1}{2}}$$

$$= \frac{\frac{12+5\sqrt{2}}{2}}{\frac{9\sqrt{2}-4}{2}}$$

$6\times\sqrt{2}\times\sqrt{2}$

$2\times\sqrt{2}\times\sqrt{2}$

$$= \frac{\sqrt{2}(6\sqrt{2}+5)}{\sqrt{2}(9-2\sqrt{2})}$$

38. If A = 10°, what is the value of

$$\frac{12\sin 3A + 5\cos (5A-5)^\circ}{9\sin\frac{9A}{2} - 4\cos (5A+10)^\circ}$$

यदि $A = 10^\circ$ है तो $\dfrac{12\sin 3A + 5\cos (5A-5)^\circ}{9\sin\frac{9A}{2} - 4\cos (5A+10)^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 18/04/2022 (Shift- 03)

(a) $\dfrac{6\sqrt{2}+5}{(9+2\sqrt{2})}$

(b) $\dfrac{6\sqrt{2}-5}{(9-2\sqrt{2})}$

(c) ✓ $\dfrac{6\sqrt{2}+5}{(9-2\sqrt{2})}$

(d) $\dfrac{(9-2\sqrt{2})}{6\sqrt{2}+5}$

$$= \frac{12 \times \frac{1}{2} + 5 \times \frac{1}{\sqrt{2}}}{9 \times \frac{1}{\sqrt{2}} - 4 \times \frac{1}{2}} = \frac{\frac{6\sqrt{2}+5}{\sqrt{2}}}{\frac{9-2\sqrt{2}}{\sqrt{2}}}$$

38. If A = 10°, what is the value of

$$\frac{12\sin 3A + 5\cos (5A-5)^\circ}{9\sin\frac{9A}{2} - 4\cos (5A+10)^\circ}$$

यदि $A = 10^\circ$ है तो $\frac{12\sin 3A + 5\cos (5A-5)^\circ}{9\sin\frac{9A}{2} - 4\cos (5A+10)^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 18/04/2022 (Shift- 03)

(a) $\frac{6\sqrt{2}+5}{(9+2\sqrt{2})}$

(b) $\frac{6\sqrt{2}-5}{(9-2\sqrt{2})}$

(c) $\frac{6\sqrt{2}+5}{(9-2\sqrt{2})}$

(d) $\frac{(9-2\sqrt{2})}{6\sqrt{2}+5}$

$\sqrt{3}\ 3\times 2 = 2\sqrt{3}\,3$

C

$\frac{\sqrt{3}}{2} = \cos\theta$

$\theta = 30^\circ$

39. If $3\tan\theta = 2\sqrt{3}\ \sin\theta$, $0^\circ < \theta < 90^\circ$, then the value of $\frac{\text{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2\theta + \tan^2 2\theta}$ is:

यदि $3\tan\theta = 2\sqrt{3}\ \sin\theta$, $0^\circ < \theta < 90^\circ$ है तो $\frac{\text{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2\theta + \tan^2 2\theta}$ का मान ज्ञात कीजिए।

SSC CGL MAINS 29/01/2022

$\frac{\text{cosec}^2 60 + \cot^2 60}{\sin^2 30 + \tan^2 60}$

$= \frac{\frac{4}{3} + \frac{1}{3}}{\frac{1}{4} + 3} = \frac{\frac{5}{3}}{\frac{13}{4}}$

$= \frac{20}{39}$

(a) $\frac{4}{13}$

(b) $\frac{20}{39}$

(c) $\frac{4}{3}$

(d) $\frac{20}{27}$

$\sin^2(2x-10)=\frac{3}{4}$

$\sin(2x-10)=\frac{\sqrt{3}}{2}=\sin 60^\circ$

$2x-10=60^\circ$

$x=35^\circ$

$$\frac{\sin^4 30^\circ+\cos^4 30^\circ}{1-2\sin^2 90^\circ\cos^2 90^\circ}=\frac{1}{16}+\frac{9}{16}$$

$$=\frac{10}{16}=\frac{5}{8}$$

40. If $4\sin^2(2x-10)^\circ = 3, 0^\circ \le (2x-10) \le 90^\circ$, then find the value of $\dfrac{\sin^4(x-5)^\circ+\cos^4(x-5)^\circ}{1-2\sin^2(3x-15)^\circ\cos^2(3x-15)^\circ}$.

यदि $4\sin^2(2x-10)^\circ = 3, 0^\circ \le (2x-10) \le 90^\circ$ है तो $\dfrac{\sin^4(x-5)^\circ+\cos^4(x-5)^\circ}{1-2\sin^2(3x-15)^\circ\cos^2(3x-15)^\circ}$ का मान ज्ञात कीजिए।

SSC CGL 17/08/2021 (Shift- 03)

(a) 1

(b) $\frac{5}{8}$

(c) $-\frac{5}{8}$

(d) -1

41. If $2\sin(3x - 15)^\circ = 1$, $0^\circ < (3x - 15) < 90^\circ$, then find the value of $\cos^2(2x + 15)^\circ + \cot^2(x + 15)^\circ$.

H.W

यदि $2\sin(3x - 15)^\circ = 1$, $0^\circ < (3x - 15) < 90^\circ$ है तो $\cos^2(2x + 15)^\circ + \cot^2(x + 15)^\circ$ का मान ज्ञात कीजिए।

SSC CGL 16/08/2021 (Shift- 02)

(a) 1

(b) $\frac{5}{2}$

(c) $-\frac{7}{2}$

(d) $\frac{7}{2}$

42. If $(2\cos A + 1)(2\cos A - 1) = 0,\ 0° < A ? 90°$, then find the value of A.

यदि $(2\cos A + 1)(2\cos A - 1) = 0,\ 0° < A ? 90°$ है तो A का मान ज्ञात कीजिए।

SSC CGL 21/04/2022 (Shift- 01)

(a) 90°
(b) 45°
(c) 30°
(d) 60°

$(2\cos A)^2 - (1)^2 = 0$

$\Rightarrow 4\cos^2 A - 1 = 0$

$\Rightarrow \cos^2 A = \frac{1}{4}$

$\Rightarrow \cos A = \frac{1}{2}$

$A = 60°$

$(a+b)(a-b) = a^2 - b^2$

$\sin(x+30^\circ) = \frac{\sqrt{3}\times\sqrt{3}}{2\sqrt{3}}$

$\sin(x+30) = \frac{\sqrt{3}}{2} = \sin 60^\circ$

$x+30 = 60$

$x = 30^\circ$

43. If $\sin (x + 30)^\circ = \frac{3}{\sqrt{12}}$, then the value of $x\ (0^\circ < x < 90^\circ)$ is : $\sqrt{4\times3}$

यदि $\sin (x + 30)^\circ = \frac{3}{\sqrt{12}}$ है तो $x\ (0^\circ < x < 90^\circ)$ का मान ज्ञात कीजिए।

SSC CHSL 26/10/2020 (Shift- 02)

(a) 60°
(b) 15°
(c) 45°
(d) 30° ✓

44. If $0° < \theta < 90°$ and $\cos^2\theta = 3(\cot^2\theta - \cos^2\theta)$, then the value of $\left(\frac{1}{2}\sec\theta + \sin\theta\right)^{-1}$ is :

यदि $0° < \theta < 90°$ तथा $\cos^2\theta = 3(\cot^2\theta - \cos^2\theta)$ है तो $\left(\frac{1}{2}\sec\theta + \sin\theta\right)^{-1}$ का मान ज्ञात कीजिए।

SSC CGL 04/06/2019 (Shift- 02)

(a) $\sqrt{3}+2$

(b) $2(2-\sqrt{3})$

(c) $2(\sqrt{3}-1)$

(d) $\sqrt{3}+1$

$\cos^2\theta = 3\cot^2\theta - 3\cos^2\theta$

$\Rightarrow 4\cos^2\theta = 3\frac{\cos^2\theta}{\sin^2\theta}$

$\sin^2\theta = \frac{3}{4}$

$\sin\theta = \frac{\sqrt{3}}{2}$

$\theta = 60°$

$\left(\frac{1}{2}\times 2 + \frac{\sqrt{3}}{2}\right)^{-1}$

$= \left(\frac{2+\sqrt{3}}{2}\right)^{-1}$

$= \frac{2}{(2+\sqrt{3})} \times \frac{(2-\sqrt{3})}{(2-\sqrt{3})}$

$= 2(2-\sqrt{3})$

$\theta = 60^\circ$

$$\frac{\frac{1}{4}}{\frac{1}{3}+\frac{3}{4}-1} = \frac{\frac{1}{4}}{\frac{4+9-12}{12}}$$

$$= \left(\frac{\frac{1}{4}}{\frac{1}{12}} = \frac{12}{4}\right) ③$$

$\tan 60^\circ + \operatorname{cosec} 60^\circ$

$$\sqrt{3}+\frac{2}{\sqrt{3}} = \frac{5}{\sqrt{3}} = \frac{5\sqrt{3}}{3}$$

45. If $\dfrac{\cos^2\theta}{\cot^2\theta+\sin^2\theta-1} = 3,\ 0^\circ < \theta < 90^\circ$, then the value of $(\tan\theta + \operatorname{cosec}\theta)$ is:

30, 45, 60

यदि $\dfrac{\cos^2\theta}{\cot^2\theta+\sin^2\theta-1} = 3,\ 0^\circ < \theta < 90^\circ$ है तो $(\tan\theta + \operatorname{cosec}\theta)$ का मान ज्ञात कीजिए।

SSC CGL 13/08/2021 (Shift- 01)

(a) $2\sqrt{3}$

(b) $\dfrac{5\sqrt{3}}{3}$

(c) $3\sqrt{3}$

(d) $\dfrac{4\sqrt{3}}{3}$

$\cos(A-B)=\cos 30^\circ$ | $\sec A=2$

$A-B=30^\circ$ | $A=60^\circ$

$\therefore 60-B=30^\circ$

$B=30^\circ$

46. If $\cos(A-B)=\frac{\sqrt{3}}{2}$ and $\sec A = 2$, $0^\circ \leq A \leq 90^\circ$, $0^\circ \leq B \leq 90^\circ$, then what is the measure of B?

यदि $\cos(A-B)=\frac{\sqrt{3}}{2}$ तथा $\sec A = 2$, $0^\circ \leq A \leq 90^\circ$, $0^\circ \leq B \leq 90^\circ$ है तो कोण B का मान ज्ञात कीजिए।

SSC CGL 11/04/2022 (Shift- 02)

(a) 60°
(b) 0°
(c) 30°
(d) 90°

$\sin\frac{(2A+B)}{2}=\frac{\sqrt{3}}{2}$ | $\cos\frac{(2A-B)}{2}=\frac{\sqrt{3}}{2}$

$\frac{2A+B}{2}=60^\circ$ | $\frac{2A-B}{2}=30^\circ$

$2A+B=120^\circ$

$+\quad 2A-B=60^\circ$

$4A=180^\circ$

$A=45^\circ$

$B=30^\circ$

$\therefore \sin 3(A-B)$

$=\sin 45^\circ$

$=\frac{1}{\sqrt{2}}$

47. **If** $\sin\left(\frac{2A+B}{2}\right)=\cos\left(\frac{2A-B}{2}\right)=\frac{\sqrt{3}}{2}$ **and** $0^\circ<\frac{2A+B}{2}<90^\circ$ **then find the value of** $\sin[3(A-B)]$.

यदि $\sin\left(\frac{2A+B}{2}\right)=\cos\left(\frac{2A-B}{2}\right)=\frac{\sqrt{3}}{2}$ तथा $0^\circ<\frac{2A+B}{2}<90^\circ$ है तो $\sin[3(A-B)]$ का मान ज्ञात कीजिए।

SSC CGL 16/08/2021 (Shift- 03)

(a) 1

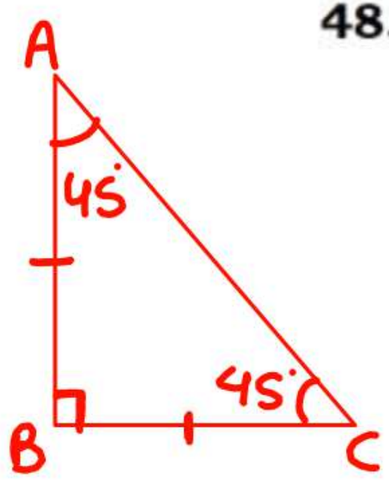
(b) $\frac{1}{\sqrt{2}}$

(c) $\frac{1}{2}$

(d) $\frac{\sqrt{3}}{2}$

$$\frac{\sin 0^\circ + \sin 90^\circ - 2\sin 90^\circ}{\cot 45^\circ + \cot 90 + \cot 45^\circ}$$

$$= \frac{0+1-2}{1+0+1} = \frac{-1}{2}$$



48. **If triangle ABC is a right isosceles triangle, right angled at B, then find the value of**

$$\frac{\sin(A-C)+\sin(A+C)-2\sin B}{\cot A+\cot B+\cot C}.$$

यदि त्रिभुज **ABC** एक समकोण समद्विबाहु त्रिभुज है, जो **B** पर समकोण है, तो $\frac{\sin(A-C)+\sin(A+C)-2\sin B}{\cot A+\cot B+\cot C}$ का मान ज्ञात कीजिए।

NTPC CBT-1 08/04/2021 (Shift-03)

(a) $\frac{1}{2}$

(b) $-\frac{1}{2}$

(c) $\frac{3}{2}$

(d) 0

49. If $\tan^2 A - 6\tan A + 9 = 0$, $0° < A < 90°$, What is the value of $6\cot A + 8\sqrt{10}\cos A$?

यदि $\tan^2 A - 6\tan A + 9 = 0$, $0° < A < 90°$ है तो $6\cot A + 8\sqrt{10}\cos A$ का मान ज्ञात कीजिए।

SSC CGL 20/04/2022 (Shift- 02)

(a) $10\sqrt{10}$ (b) $8\sqrt{10}$

(c) 10 (d) 14

$t^2 - 6t + 9 = 0$

$\Rightarrow t^2 - 3t - 3t + 9 = 0$

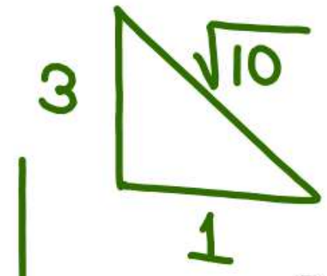
$\Rightarrow t(t-3) - 3(t-3) = 0$

$\Rightarrow (t-3)(t-3) = 0$

$t - 3 = 0$

$t = 3$

$\tan\theta = \frac{3}{1} = \frac{P}{B}$



$\cot A = \frac{1}{3}$, $\cos A = \frac{1}{\sqrt{10}}$

Ans $= 6 \times \frac{1}{3} + 8\sqrt{10} \times \frac{1}{\sqrt{10}}$

$= 2 + 8$

$= 10$

$t^2 - 6t + 9 = 0$

$+3 \quad +3$

$1 \quad 1$

$\tan\theta = \frac{3}{1}$ → P, 1 → B

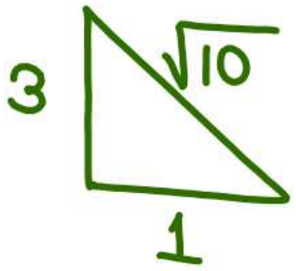
49. If $\tan^2 A - 6 \tan A + 9 = 0$, $0° < A < 90°$, What is the value of $6\cot A + 8\sqrt{10}\cos A$?

यदि $\tan^2 A - 6 \tan A + 9 = 0$, $0° < A < 90°$ है तो $6\cot A + 8\sqrt{10}\cos A$ का मान ज्ञात कीजिए।

SSC CGL 20/04/2022 (Shift- 02)

(a) $10\sqrt{10}$ (b) $8\sqrt{10}$

(c) 10 (d) 14



$\text{Ans} = 6 \times \frac{1}{3} + 8\sqrt{10} \times \frac{1}{\sqrt{10}}$

$= 2 + 8$

$= 10$

50. If $3\sec\theta + 4\cos\theta - 4\sqrt{3} = 0$ where θ is an acute angle then the value of θ is:

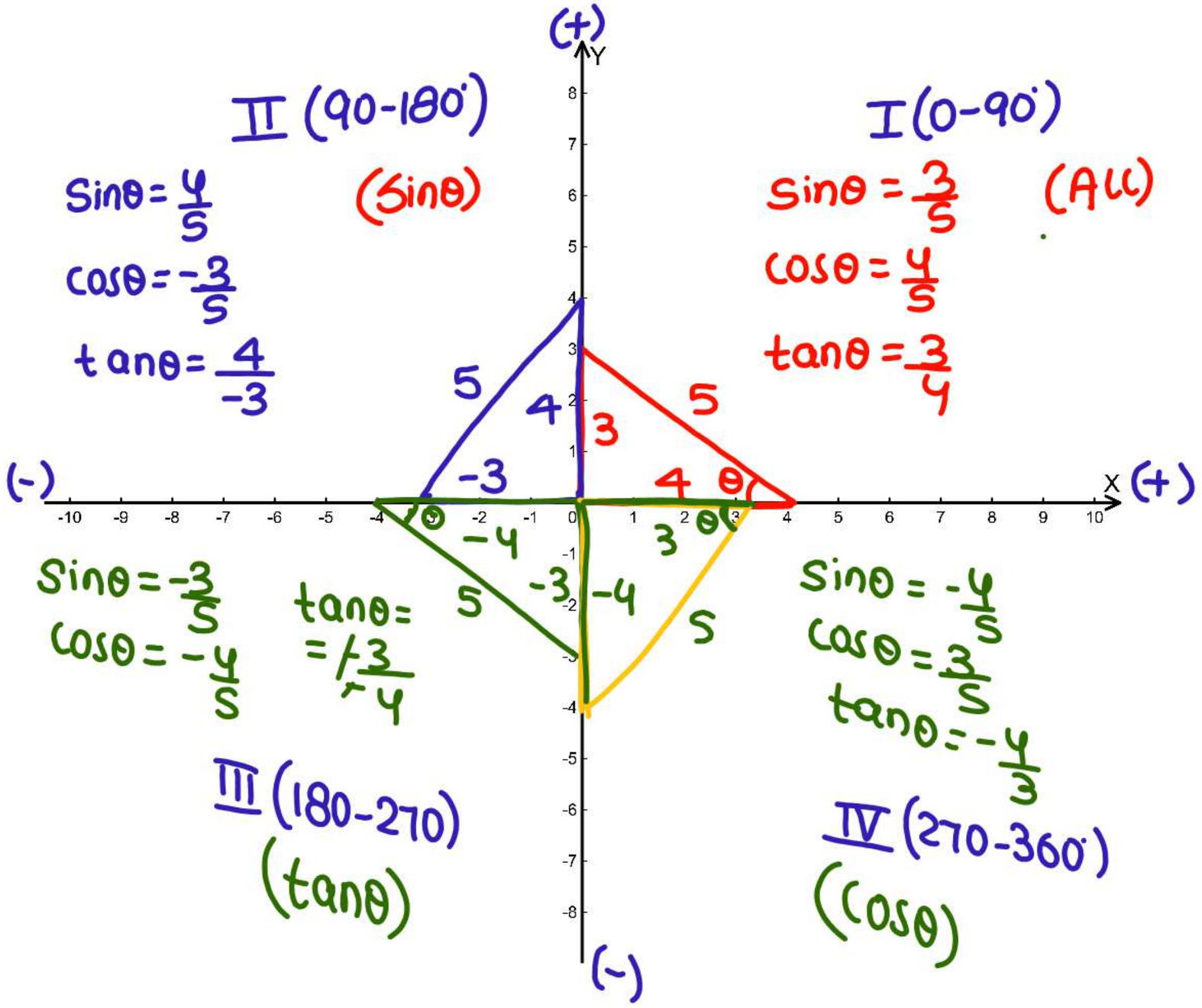
यदि $3\sec\theta + 4\cos\theta - 4\sqrt{3} = 0$ जहां θ एक न्यूनकोण है तो θ का मान ज्ञात कीजिए।

SSC CGL 13/08/2021 (Shift- 03)

(a) 20°
(b) 30°
(c) 60°
(d) 45°

$$3 \times \frac{2}{\sqrt{3}} + 4 \times \frac{\sqrt{3}}{2} - 4\sqrt{3}$$

$$4\sqrt{3} - 4\sqrt{3} = 0$$


(+)
Y
X
(+)
(-)
(-)
II (90-180°)
(Sinθ)
Sinθ = 4/5
Cosθ = -3/5
tanθ = 4/-3
I (0-90°)
(All)
Sinθ = 3/5
Cosθ = 4/5
tanθ = 3/4
III (180-270)
(tanθ)
Sinθ = -3/5
Cosθ = -4/5
tanθ = -3/-4
IV (270-360°)
(Cosθ)
Sinθ = -4/5
Cosθ = 3/5
tanθ = -4/3
5
4
-3
3
5
-4
-3
5
3
4
θ
θ
θ

1st Quadrant

$\sin(-\theta) = -\sin\theta$

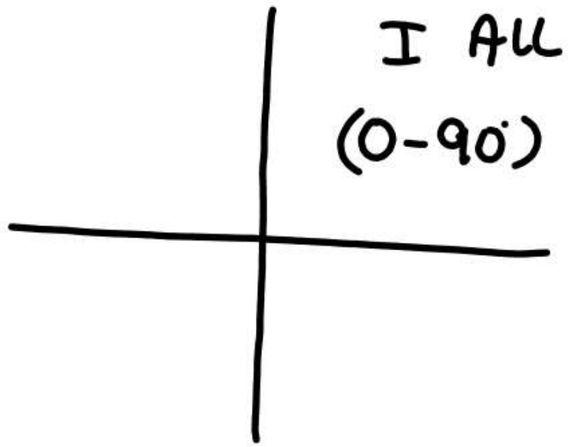
$\cos(-\theta) = \cos\theta$

$\tan(-\theta) = -\tan\theta$

$\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$

$\sec(-\theta) = \sec\theta$

$\cot(-\theta) = -\cot\theta$



$\therefore$

Note:

$90 \times 1 = 90$
$90 \times 3 = 270$
$90 \times \text{odd} \rightsquigarrow$

$\sin\theta \rightleftharpoons \cos\theta$
$\operatorname{cosec}\theta \rightleftharpoons \sec\theta$
$\tan\theta \rightleftharpoons \cot\theta$

$90 \times 2 = 180$
$90 \times 4 = 360$
$90 \times 6 = 540$

No change

1st Quadrant

$\sin(90-\theta) = +\cos\theta$

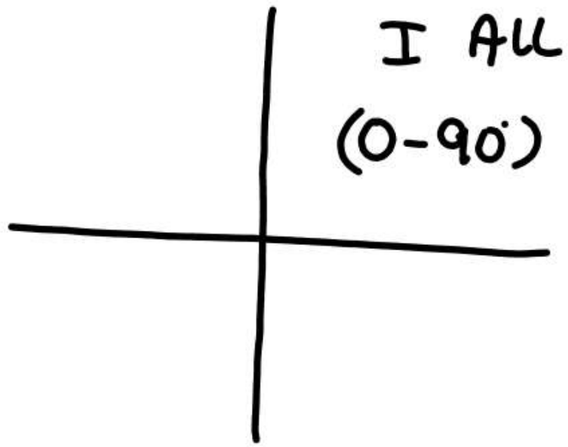
$\cos(90-\theta) = +\sin\theta$

$\tan(90-\theta) = +\cot\theta$

$\operatorname{cosec}(90-\theta) = +\sec\theta$

$\sec(90-\theta) = +\operatorname{cosec}\theta$

$\cot(90-\theta) = +\tan\theta$



2nd Quadrant

$\sin(90+\theta) = +\cos\theta$

$\cos(90+\theta) = -\sin\theta$

$\tan(90+\theta) = -\cot\theta$

$\operatorname{cosec}(90+\theta) = +\sec\theta$

$\sec(90+\theta) = -\operatorname{cosec}\theta$

$\cot(90+\theta) = -\tan\theta$

$\sin(180-\theta) = +\sin\theta$

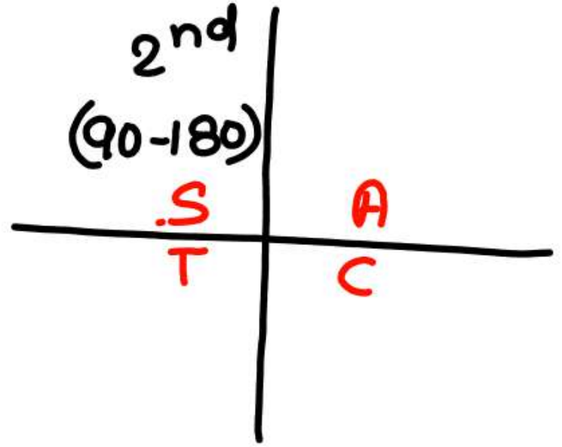
$\cos(180-\theta) = -\cos\theta$

$\tan(180-\theta) = -\tan\theta$

$\operatorname{cosec}(180-\theta) = +\operatorname{cosec}\theta$

$\sec(180-\theta) = -\sec\theta$

$\cot(180-\theta) = -\cot\theta$



3rd Quadrant

$\sin(180+\theta) = -\sin\theta$

$\cos(180+\theta) = -\cos\theta$

$\tan(180+\theta) = +\tan\theta$

$\operatorname{cosec}(180+\theta) = -\operatorname{cosec}\theta$

$\sec(180+\theta) = -\sec\theta$

$\cot(180+\theta) = +\cot\theta$

$\sin(270-\theta) = -\cos\theta$

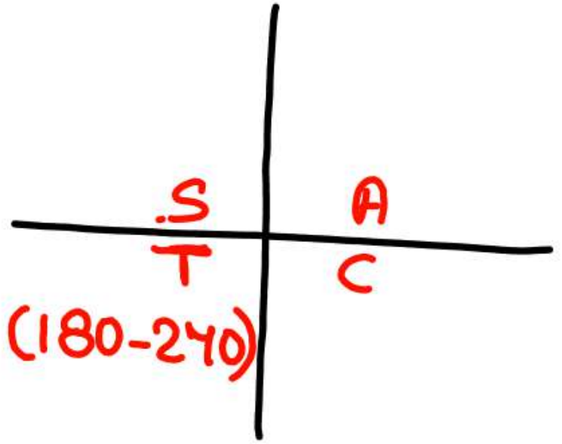
$\cos(270-\theta) = -\sin\theta$

$\tan(270-\theta) = +\cot\theta$

$\operatorname{cosec}(270-\theta) = -\sec\theta$

$\sec(270-\theta) = -\operatorname{cosec}\theta$

$\cot(270-\theta) = +\tan\theta$



4th Quadrant

$\sin(270+\theta) = -\cos\theta$

$\cos(270+\theta) = +\sin\theta$

$\tan(270+\theta) = -\cot\theta$

$\operatorname{cosec}(270+\theta) = -\sec\theta$

$\sec(270+\theta) = +\operatorname{cosec}\theta$

$\cot(270+\theta) = -\tan\theta$

$\frac{270}{90}$ 3

$\sin(360-\theta) = -\sin\theta$

$\cos(360-\theta) = +\cos\theta$

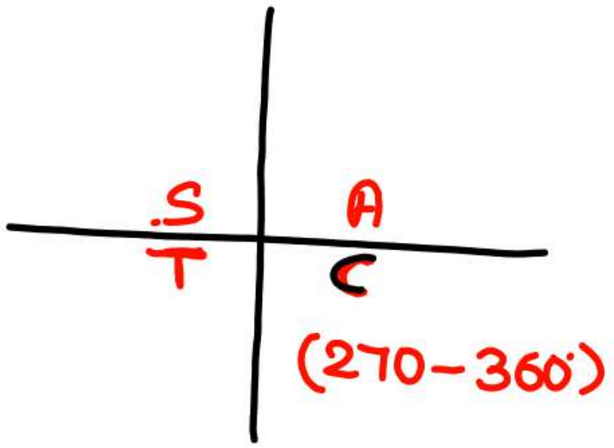
$\tan(360-\theta) = -\tan\theta$

$\operatorname{cosec}(360-\theta) = -\operatorname{cosec}\theta$

$\sec(360-\theta) = +\sec\theta$

$\cot(360-\theta) = -\cot\theta$

$\frac{360}{90}$ 4



51. Sin (– A) = ?

SSC CHSL 24/05/2022 (Shift- 02)

(a) cos A
(b) – cos A
(c) – sin A
(d) sin A

52. $\tan(\pi + \theta) = ?$

SSC CHSL 08/06/2022 (Shift-01)

(a) $\sec\theta$ (b) $\operatorname{cosec}\theta$

(c) $\cot\theta$ (d) $\tan\theta$

$\tan(180+\theta)$

$= +\tan\theta$

S	A
90-180	0-90
180-270	
T	C

52. $\tan(\pi + \theta) = ?$

SSC CHSL 08/06/2022 (Shift-01)

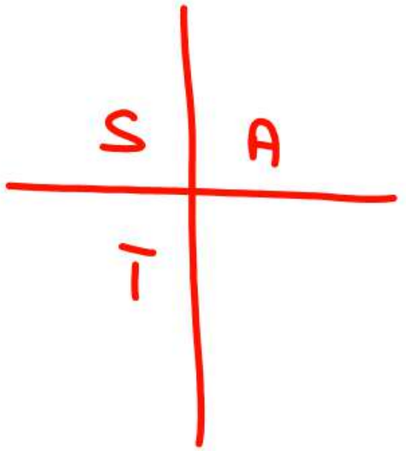
(a) $\sec\theta$ (b) $\operatorname{cosec}\theta$

(c) $\cot\theta$ (d) $\tan\theta$

$\tan(180+\theta)$

$= \tan(90\times2+\theta)$

$= +\tan\theta$



53. **Find the exact value of sin150°.**

sin150° का सटीक मान ज्ञात कीजिए।

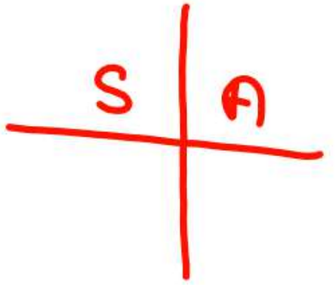
SSC CGL 12/09/2024 (Shift-03)

(a) 1.5
(b) 0.5
(c) 0.75
(d) –0.5

$= \sin(90 \times 1 + 60)$

$= +\cos 60°$

$= \frac{1}{2} = 0.5$



54. What is the value of tan 570°

tan 570° का मान कितना होगा?

SSC CHSL 04/07/2024 (Shift-02)

(a) $-\frac{1}{\sqrt{3}}$

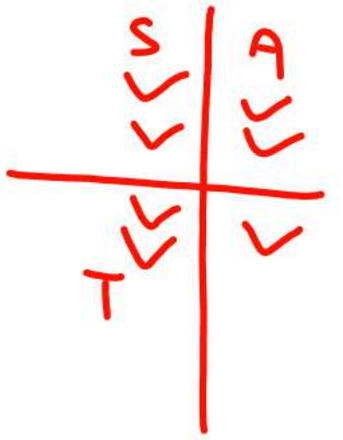
(b) $-\sqrt{3}$

(c) $\frac{1}{\sqrt{3}}$

(d) $\sqrt{3}$

$\tan(90\times6+30)$

$= +\tan 30^\circ = \frac{1}{\sqrt{3}}$



55. **What is the value of sin(–405°)?**

sin(–405°) का मान ज्ञात करें।

SSC SSC 20/07/2023 (Shift-03)

(a) $\frac{\sqrt{5}}{2}$

(b) $\frac{1}{2}$

(c) $\frac{-1}{2}$

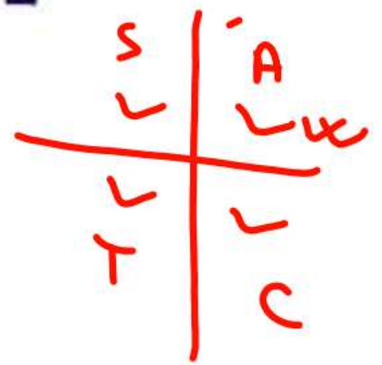
(d) $\frac{-1}{\sqrt{2}}$

$= -\sin 405^\circ$

$= -[\sin(90\times4+45)]$

$= -[+\sin 45^\circ]$

$= -\frac{1}{\sqrt{2}}$



56. **Find the value of tan(–1125°).**

tan(–1125°) का मान ज्ञात कीजिए।

SSC CGL 25/07/2023 (Shift-2)

(a) 1

(b) $\frac{1}{2}$

(c) –1

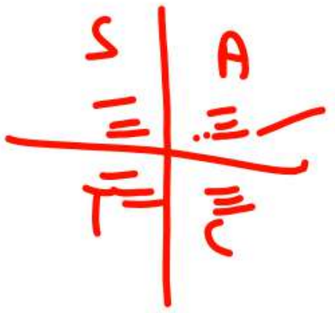
(d) 0

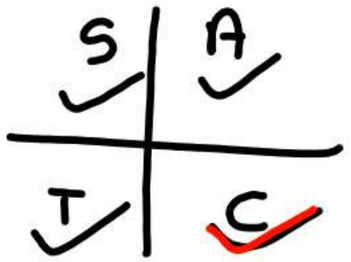
$= -\tan(1125^\circ)$

$= -\tan(90^\circ \times 12 + 45^\circ)$

$= -\{+\tan 45^\circ\}$

$= -1$





57. Find the value of tan4384° + cot6814° = ?

tan4384° + cot6814° का मान ज्ञात कीजिए।

SSC CGL 26/07/2023 (Shift-3)

(a) −1 **(b) 2**

(c) 0 **(d) 1**

$= \tan(90\times48+64) + \cot(90\times75+64)$

$= +\tan 64^\circ \qquad -\tan 64^\circ$

$= 0$

S	All
T.	C

58. Find the value of
$\cos570^\circ \sin510^\circ + \sin(-330^\circ)\cos(-390^\circ)$
का मान ज्ञात कीजिए।

(a) 1 (b) 0
(c) 2 (d) NOT

$= \cos(90\times6+30^\circ)\sin(90\times5+60^\circ) - \sin(90\times3+60^\circ)\cos(90\times4+30^\circ)$

$= (-\cos30^\circ)(+\cos60^\circ) - [(-\cos60^\circ)(+\cos30^\circ)]$

$= -\frac{\sqrt3}{2}\times\frac{1}{2}+\frac{1}{2}\times\frac{\sqrt3}{2}$

$= -\frac{\sqrt3}{4}+\frac{\sqrt3}{4}$

$= 0$

$\pi = 180^\circ$

$\therefore \frac{\pi}{2} = \frac{180}{2} = 90^\circ$

$\therefore \frac{\pi}{3} = \frac{180}{3} = 60^\circ$

$\therefore \frac{\pi}{4} = \frac{180}{4} = 45^\circ$

$\therefore \frac{\pi}{6} = \frac{180}{6} = 30^\circ$

(Sin 315)(sin45°)(sin135°)(sin225)

= Sin(90x3+45°)x $\frac{1}{\sqrt{2}}$ x sin(90+45°)·sin(90x2+45°)

= −cos45°x$\frac{1}{\sqrt{2}}$ x cos45°x(− sin45°)

= $+\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{4}$

59. **Find is the value of $\sin\frac{7\pi}{4}$ $\sin\frac{\pi}{4}$ $\sin\frac{3\pi}{4}$ $\sin\frac{5\pi}{4}$.**

$\sin\frac{7\pi}{4}$ $\sin\frac{\pi}{4}$ $\sin\frac{3\pi}{4}$ $\sin\frac{5\pi}{4}$ का मान ज्ञात करें।

RRB JE 22/05/2019 (Shift-02)

(a) $\frac{1}{4}$ (b) $\frac{3}{16}$

(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

S A T C

$\Rightarrow \cot\theta = \frac{2}{\sqrt{3}}$

$2\sqrt{3} \times \frac{\sqrt{3}}{2} + 1$

$= 4$

60. **If $\tan(90° - \theta) = \frac{2}{\sqrt{3}}$, then the value of $2\sqrt{3}\tan\theta + 1$ is:**

यदि $\tan(90° - \theta) = \frac{2}{\sqrt{3}}$, तो $2\sqrt{3}\tan\theta + 1$ का मान ज्ञात कीजिए।

SSC CHSL 11/08/2023 (Shift-2)

(a) 4
(b) 5
(c) 3
(d) 6

61. What is the simplified value of $\cos^2(90°-\theta) - \left[\frac{\{\cos(90°-\theta)\cos\theta\}}{\cot\theta}\right]$?/का सरलीकृत मान क्या है?

SSC CHSL 31/05/2022 (Shift- 3)

(a) 4
(b) 2
(c) 0
(d) 1

$= \sin^2\theta - \frac{\sin\theta\cos\theta \times \sin\theta}{\cos\theta}$

$= \sin^2\theta - \sin^2\theta$

$= 0$

$\sin^2\theta+\cos^2\theta=1$

$\Rightarrow \cos^2\theta-1=-\sin^2\theta$

62. Find the value of

$$\frac{\sin(90^\circ-\theta)\cos(90^\circ-\theta)\cot(90^\circ-\theta)}{\cos^2\theta-1}$$

का मान ज्ञात कीजिए।

Group D 19/09/2022 (Shift-01)

(a) $\tan\theta$ (b) 0

(c) -1 (d) $2\sin\theta\cos\theta$

$$= \frac{\cos\theta\sin\theta\cdot\frac{\sin\theta}{\cos\theta}}{-\sin^2\theta}$$

$= -1$

- $\sin C + \sin D = 2\sin\frac{(C+D)}{2}\cos\frac{(C-D)}{2}$
- $\sin C - \sin D = 2\cos\frac{(C+D)}{2}\sin\frac{(C-D)}{2}$
- $\cos C + \cos D = 2\cos\frac{(C+D)}{2}\cos\frac{(C-D)}{2}$
- $\cos C - \cos D = 2\sin\frac{(C+D)}{2}\sin\frac{(D-C)}{2}$

$\sin 20^\circ = \cos 70^\circ$

$\sin 10^\circ = \cos 80^\circ$

63. **The following expression is equal to/नीचे दिया गया व्यंजक बराबर है।**

$\cot 85^\circ + \cos 75^\circ$

SSC CGL 20/07/2023 (Shift- 02)

(a) $\tan 85^\circ + \sin 75^\circ$ **(b) $\tan 85^\circ - \sin 75^\circ$**

(c) $\tan 5^\circ + \sin 15^\circ$ **(d) $\tan 5^\circ - \sin 15^\circ$**

$\tan 5^\circ + \sin 15^\circ$

64. The value of sin 73° + cos137° is

sin73° + cos137° का मान क्या होगा?

SSC CHSL 10/06/2022 (Shift- 3)

(a) sin 13°
(b) cos 13°
(c) cos 18°
(d) sin 18°

$\sin 73^\circ + \cos(90+47^\circ)$

$= \sin 73^\circ - \sin 47^\circ$

$= 2\cos\frac{120}{2}\sin\frac{26}{2}$

$= 2\cos 60 \sin 13$

$= 2 \times \frac{1}{2}\sin 13$

a) $\cos(90\times2+90^\circ) = -\cos 90^\circ = 0$ 65.

X b) $\cos(90\times4+90^\circ) = +$

c) $\cos(90\times1+60^\circ) = -\sin 60^\circ = -\frac{\sqrt{3}}{2}$

d) $\cos(90\times2+60^\circ) = -\cos 60^\circ = -\frac{1}{2}$

Find θ, if $\cos\theta = -\frac{\sqrt{3}}{2}$

यदि $\cos\theta = -\frac{\sqrt{3}}{2}$ है, तो θ का मान ज्ञात करें।

SSC CHSL 02/06/2022 (Shift- 2)

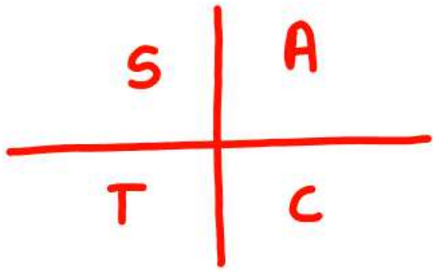
(a) $\frac{3\pi}{2} = 270^\circ$

(b) $\frac{5\pi}{2} = 450^\circ$

(c) $\frac{5\pi}{6} = 150^\circ$

(d) $\frac{4\pi}{3} = 240^\circ$

S	A
T	C



66. The value of $\sin^2\frac{2\pi}{3} + \cos^2\frac{5\pi}{6} - \tan^2\frac{3\pi}{4}$ is:

$\sin^2\frac{2\pi}{3} + \cos^2\frac{5\pi}{6} - \tan^2\frac{3\pi}{4}$ का मान क्या है?

SSC Phase X 03/08/2022 (Shift- 02)

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) 4 (d) 2

$= \sin^2 120 + \cos^2 150 - \tan^2 135$

$= \sin^2(90+30) + \cos^2(90+60) - \tan^2(90+45)$

$= \cos^2 30 + \sin^2 60 - \cot^2 45$

$= \frac{3}{4} + \frac{3}{4} - 1 = \frac{2}{4} = \frac{1}{2}$

$$\sec x \sqrt{\frac{-\cos x \cos x}{\sin x(-\sin x)}}$$

$$= \frac{1}{\cancel{\cos x}} \times \frac{\cancel{\cos x}}{\sin x} = \operatorname{cosec} x$$

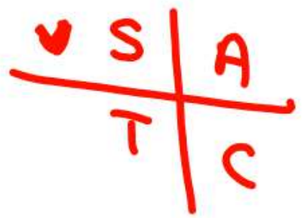
67. Simplify. $\dfrac{1}{\cos x}\sqrt{\dfrac{\cos(\pi + x)\cos(-x)}{\sin(\pi - x)\cos\left(\dfrac{\pi}{2} + x\right)}}$

$\dfrac{1}{\cos x}\sqrt{\dfrac{\cos(\pi + x)\cos(-x)}{\sin(\pi - x)\cos\left(\dfrac{\pi}{2} + x\right)}}$ सरल करें

SSC Phase X 04/08/2022 (Shift- 03)

(a) $\tan x$ (b) $\cot x$

(c) $\sec x$ (d) $\operatorname{cosec} x$



$x = 30°$

68. If $\cos x = -\frac{\sqrt{3}}{2}$ and $\pi < x < \frac{3\pi}{2}$, then the value of $2\cot^2 x + 3\sec^2 x$ is :

SSC CHSL 08/07/2019 (Shift- 02)

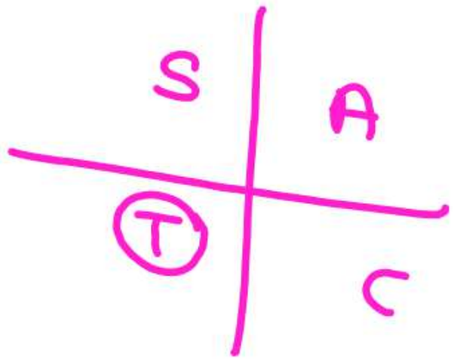
(a) 10
(b) 4
(c) 8
(d) 16

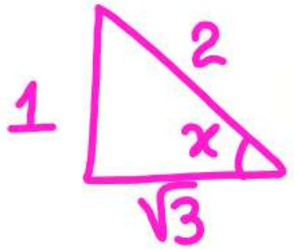
$= 2\cot^2 x + 3\sec^2 x$

$= 2\cot^2 30 + 3\sec^2 30$

$= 2 \times 3 + 3 \times \frac{4}{3}$

$= 10$





68. If $\cos x = -\frac{\sqrt{3}}{2}$ and $\pi < x < \frac{3\pi}{2}$, then the value of $2\cot^2 x + 3\sec^2 x$ is :

SSC CHSL 08/07/2019 (Shift- 02)

(a) 10 (b) 4
(c) 8 (d) 16

B

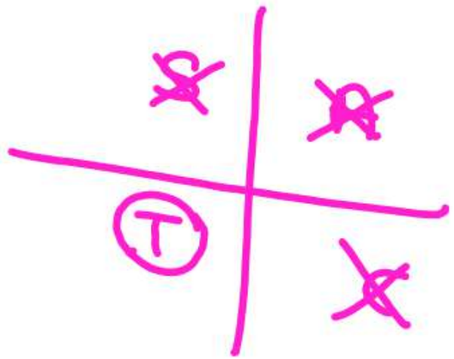
3rd

H

$= 2\cot^2 x + 3\sec^2 x$

$= 2 \times 3 + 3 \times 4$

$= 10$



69. Find x if $\cos x = -\frac{1}{2}$, $\left(\frac{\pi}{2} < x < \pi\right)$

$90 < x < 180^\circ$

SSC CHSL 15/10/2020 (Shift-03)

(a) $\frac{3\pi}{2} = 270$

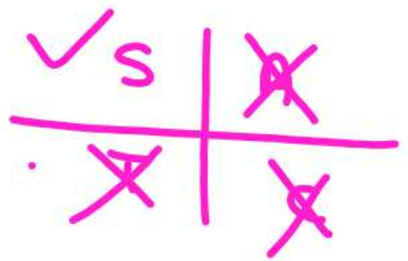
(b) $\frac{2\pi}{3} = 120^\circ$

(c) $\frac{5\pi}{6} = 150$

(d) $\frac{4\pi}{3} = 240^\circ$

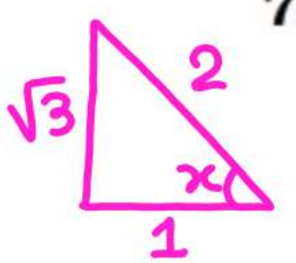
(b) $\cos(120^\circ) = \cos(90+30^\circ)$
$= -\sin 30^\circ = -\frac{1}{2}$

(c) $\cos(150) = \cos(90+60)$
$= -\sin 60$
$= -\frac{\sqrt{3}}{2}$



$\cos x = \frac{1}{2}$ → B, 2 → H

$\tan x = +\frac{\sqrt{3}}{1}$



70. If $\cos x = -\frac{1}{2}$, x lies in third quadrant, then $\tan x$ = ?

यदि $\cos x = -\frac{1}{2}$, x तीसरे चतुर्थांश में स्थित है, तो $\tan x$ = ?

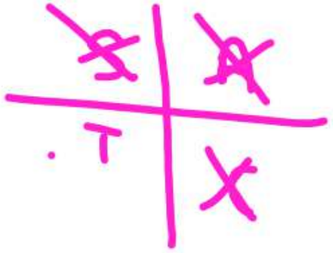
SSC CGL 17/07/2023 (Shift-03)

(a) $\sqrt{3}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{2}{\sqrt{3}}$

(d) $\frac{1}{\sqrt{3}}$



$= \sin^2\phi - \sin^2\phi + 1 \times \cos^2\phi$

71. Find the value of the following?

$$\cos^2(270^\circ - \phi) - \sin^2(180^\circ - \phi) + \sin^2\left(\frac{\pi}{2}\right)\sin^2(270^\circ - \phi).$$

निम्न का मान ज्ञात कीजिए?

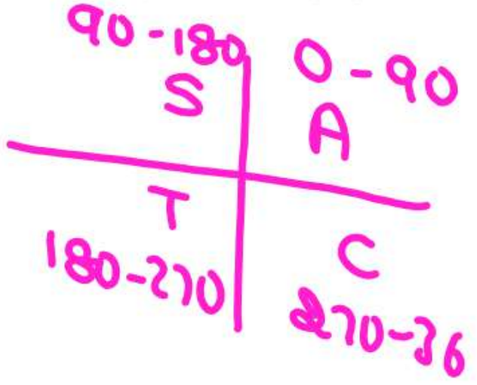
RRB NTPC 08/01/2021 (Shift-01)

(a) $\cos^2\phi$

(b) $\sin^2\left(\frac{\pi}{2}\right)$

(c) $\sin^2(\phi) - 1$

(d) $\sin^2(\phi)$



72. What is the value of
[cos(90° + A) ÷ sec(270° – A)] + [sin(270° + A) ÷ cosec(630° – A)]
का मान ज्ञात कीजिए।

H·w

(a) 3 sec A
(b) tanA secA
(c) 0
(d) 1

73. If $\sec\theta + \tan\theta = 2+\sqrt{5}$, then the value of $\sin\theta + \cos\theta$ is :

यदि $\sec\theta + \tan\theta = 2+\sqrt{5}$ है तो $\sin\theta + \cos\theta$ का मान ज्ञात कीजिए।

(a) $\frac{3}{\sqrt{5}}$ (b) $\sqrt{5}$

(c) $\frac{7}{\sqrt{5}}$ (d) $\frac{1}{\sqrt{5}}$

Lotiy

$\sec\theta + \tan\theta = p$

$\sec\theta = \frac{p^2+1}{2p}$

Sheet 2/3/4